



# Synergizing blockchain and AI to fortify IoT security: a comprehensive review

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## Abstract

The relentless growth of connected devices is transforming industrial, urban and domestic environments, yet it also expands the attack surface for distributed denial of service (DDoS), unauthorized access and data manipulation. Centralized security architectures struggle to cope with the scale and heterogeneity of the Internet of Things, creating single points of failure and privacy risks. This review takes a close look at how blockchain and artificial intelligence (AI) can work together to solve these problems. Blockchain plays an important role in decentralizing trust, maintaining data integrity, and enabling transparent audit trails. AI subfields such as machine learning (ML), deep learning (DL), reinforcement learning (RL), and multi-agent systems (MAS) enhance these benefits. They enable real-time anomaly detection, predictive analytics, and adaptive policy control. A seven axis Blockchain–AI Security Integration Schema (BASIS) is proposed to classify solutions by security objectives, intelligence modalities, trust primitives, deployment choices, scalability techniques, privacy controls and interoperability mechanisms. In this study also review Layer-2 consensus protocols, federated learning and lightweight deep learning models that address energy and computational constraints. Case studies from supply chains, healthcare and smart grids illustrate the benefits and limitations of current deployments. The evidence suggests that while AI improves the accuracy and responsiveness of threat detection, blockchain offers tamper-proof data provenance. However, there are still issues in achieving scalability, reducing computational overhead, and striking a balance between auditability and privacy. Hybrid on-chain/off-chain architectures, quantum-safe cryptography, and standardized frameworks to guarantee adoption and interoperability are some future research avenues.

**Keywords** Blockchain · Artificial intelligence · IoT security · Scalability · Privacy

# 1 Introduction

The continuously growth of IoT devices is reshaping industries with number of connected IoT devices to grow 13% by end of 2024 (Sinha 2024). This growth creates new opportunities for innovation in fields such as smart cities, industrial processes, and the healthcare sector. However it also raises security issues, making IoT networks vulnerable to DDoS attacks, identity theft, and illegal access (Yu et al. 2018; Tang et al. 2022). Traditional centralized security models face difficulties with the decentralized and scalable nature of IoT systems which often operate across highly dispersed networks. Such a configuration raises issues of authentication, data integrity, and secure communication. The challenges are even aggravated by the fact that many IoT devices are resource-scarce, thus their ability to employ stiff security measures is limited and they are vulnerable to a variety of attacks (Attkan and Ranga 2022). Thus, new approaches are needed to protect IoT systems at a large scale. Machine learning and deep learning under the broad umbrella of AI have also found a lot of usefulness in improving IoT security in real time. With a large amount of data produced by IoT devices AI is able to identify the abnormalities, invasions of security and even adapt to new forms of threats on its own (Qu et al. 2022; Abed and Anupam 2023). Modern machine learning techniques enhance the implementation of smart Intrusion Detection Systems (IDS), that constantly supervises the traffic of IoT networks, as well as perform appropriate actions for counteracting cybersecurity threats. Simultaneously blockchain offers a Decentralized, immutable, and transparent framework that enhances the security of IoT devices by documenting transactions and interactions among devices, while also securing data exchange and communication within IoT networks (Sharma and Dhiman 2025). The consensus mechanisms of blockchain reduce reliance on centralised managing parties, reducing the number of bottlenecks and increasing the level of confidence between IoT devices. The integration of AI and blockchain yields a robust security framework that enhances data integrity and improves threat detection and mitigation in IoT networks (Alharbi et al. 2022).

Table 1 compares traditional IoT security techniques with blockchain-AI approaches and highlights enhanced scalability, privacy and decentralization with blockchain and AI. Traditional methods are relying on centralized frameworks, face limitations in scalability and real-time threat detection while blockchain and AI provide secure, adaptive and decentralized solutions for IoT vulnerabilities.

## 1.1 Motivation

The rapid expansion of IoT systems is transforming sectors like healthcare, smart grids, and transportation, but it also heightens exposure to security threats such as botnet attacks, identity spoofing, and data manipulation. Traditional centralized architectures struggle in these distributed, resource-constrained environments, leaving them vulnerable to advanced threats (Fadi et al. 2022; Dritsas and Trigka 2024). Blockchain offers decentralized trust with tamper-resistant ledgers, immutable histories, and secure authentication, while AI enables adaptive, real-time threat detection, anomaly recognition, and predictive analytics. Together, they can ensure data integrity, provenance, and rapid mitigation, but practical integration must overcome challenges of scalability, latency, energy efficiency, and interoperability through strategies like lightweight AI models, federated learning, and Layer-2

**Table 1** Comparison of IoT security approaches in different sectors

| References             | Sector                | IoT use-cases                                                         | Security gaps                                                                        | Conventional mitigation approaches                                | Observed weaknesses in conventional methods                                                                                                                          | Notable blockchain-AI implementations                                                                                                                                       | Research and deployment challenges                                                                                                    |
|------------------------|-----------------------|-----------------------------------------------------------------------|--------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Ali et al. (2025)      | Healthcare            | Remote patient monitoring, wearable health devices, medical equipment | Data breaches due to constrained devices and centralized servers; ransomware attacks | Encryption protocols, firewalls, two-factor authentication        | Static encryption cannot adapt to evolving attack vectors; firewalls fail to block insider or advanced threats; 2FA is bypassed by phishing and malware              | Hyperledger-based EHR sharing systems with smart contracts improve traceability; swarm-based blockchain for secure EHR transfer; AI anomaly detection for vitals monitoring | High network latency and synchronization delays in blockchain-AI systems; interoperability across hospitals still limited             |
| Hussain (2024)         | Smart Cities          | Smart traffic management, utility monitoring, urban surveillance      | Distributed denial-of-service (DDoS), spoofing, unauthorized data manipulation       | Centralized access control, VPNs, encryption                      | Centralized schemes create single points of failure and have poor scalability; access control misses insider threats; VPNs become bottlenecks in large city networks | Public-ledger systems store sensor data immutably and provide decentralized authentication; AI algorithms predict congestion and detect attacks                             | Blockchain scalability and energy consumption remain issues; diverse urban devices lack unified data formats                          |
| Menon et al. (2025)    | Industrial Automation | Predictive maintenance, robotics, factory automation                  | System downtime, intellectual property theft, unauthorized reconfiguration           | Intrusion detection systems, patch management, perimeter security | IDS miss zero-day attacks; patches cannot protect legacy equipment; perimeter controls overlook insider or wireless threats                                          | Blockchain ensures tamper-proof logs and machine-to-machine authentication; AI and reinforcement learning models detect anomalies and optimize resource use                 | Throughput limitations of blockchain hinder high-speed industrial operations; integration complexity across heterogeneous controllers |
| Vardakis et al. (2024) | Smart Homes           | Home automation systems, smart locks, cameras, thermostats            | Unauthorized access, data breaches, device tampering                                 | Password management, network segmentation, firmware updates       | Users often leave default passwords; segmentation is complex for non-experts; inconsistent updates leave devices exposed                                             | Blockchain secures device identities and logs activities; AI monitors unusual device behavior for intrusion detection                                                       | High adoption costs and usability hurdles; limited standards for interoperability across household brands                             |
| Avci and Koca (2024)   | Transportation        | Connected vehicles, autonomous systems, smart navigation              | Remote hacking, GPS spoofing, data privacy breaches                                  | Firewalls, antivirus software, static encryption, physical tokens | Firewalls fail to adapt to emerging threats; encryption fails if keys are compromised; tokens can be lost; real-time anomaly detection is lacking                    | Permissioned ledgers manage vehicle identities and ensure tamper-proof journey logs; AI (including RL) monitors vehicular telemetry for anomalies                           | On-chain transaction finality introduces latency; training RL agents for safety-critical tasks remains challenging                    |

blockchain solutions (CR et al. 2024; Olawale and Ebadinezhad 2024). Figure 1 outlines the problems resulting from IoT security and the integration of Blockchain with AI as solution.

In blockchain-enabled IoT networks, AI can process data at the edge, reducing the need for centralized computation. This distributed approach ensures faster data processing and minimizes latency in security decision-making (Adam et al. 2024). For securing IoT devices, blockchain can solve the problem of IoT devices' communication by use of consensus. AI here improves this process by identifying any irregular communication or an attack as the network connects only verified devices (Chithanuru and Ramaiah 2023). Blockchain integration makes AI evolve to easily detect an anomaly since the ledger created cannot be altered in any way, thus ensuring the data's correctness across IoT devices. This allows the building of sound anomaly detection mechanisms that work in real time so that any threat can be detected early (Singh et al. 2020a).

## 2 Research questions

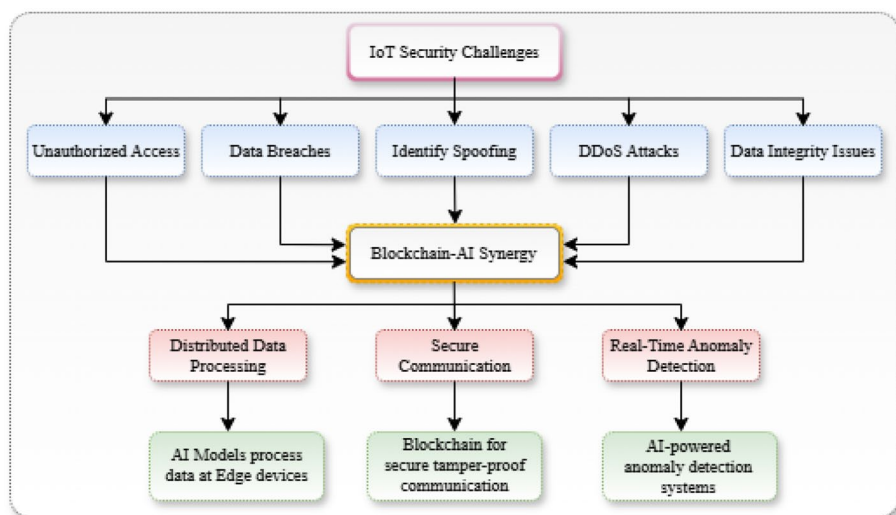
As various connected devices continue to be incorporated within various critical infrastructures, the protection of these networks has become extremely necessary. This kind of security paradigm does not work in IoT environments as they are highly dynamic and have a decentralized architecture. The following questions provide the foundation for analyzing the state of the art and identifying future research opportunities.

- (i) How can Blockchain enhance the security of IoT networks?
- (ii) How does AI contribute to threat detection and decision-making in IoT security?
- (iii) How effective is the integration of blockchain and AI in enhancing IoT security, particularly in addressing challenges such as scalability, computational complexity, and privacy?
- (iv) What are the key challenges in integrating Blockchain and AI for IoT security?

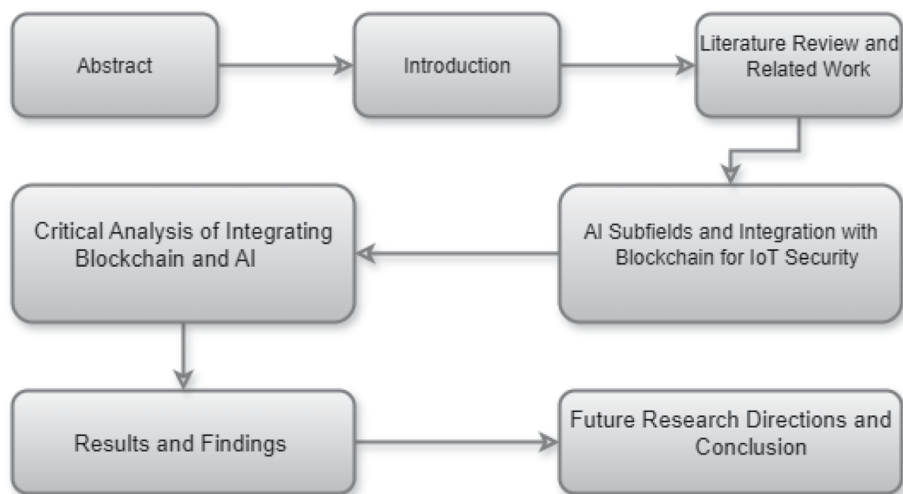
Followings are the key contributions of this paper:

- A structured review of blockchain and AI integration for IoT security, emphasising critical analysis rather than descriptive summary.
- Proposes BASIS a seven-axis Blockchain–AI Security Integration Schema that unifies blockchain primitives, AI modalities, deployment strategies, scalability methods, privacy controls, and interoperability factors into a single operational framework for IoT security.
- Real-world case studies demonstrating the benefits and limitations of deployed systems, with lessons for practitioners.
- A dedicated discussion of energy-efficient AI methods and consensus protocols (Green AI), and an analysis of standardization and interoperability challenges.
- Explicit answers to the research questions and a forward-looking agenda outlining future research directions.

The structure of this study is divided into six parts, as demonstrated in Fig. 2. The introduction part focuses on the need to combine blockchain and AI in solving IoT security



**Fig. 1** AI and Blockchain integration for addressing IoT security challenges



**Fig. 2** Various sections of the review paper

problems. Section two focuses on core issues like the improvement of security, issues related to scalability, computational complexity and privacy and data integrity of IoT systems. Section three focuses on AI subfields like ML—anomaly detection, DL—real-time threat, Response, RL—AI Decision making, and MAS—AI Secure distributed systems. Section four describes solutions such as quantum resistant blockchain algorithms, shallow AI models and both on-chain and off-chain computation modes as a way of eradicating current hindrances. In the fifth part include results and findings. The last part summarises the conclusions and future directions, outlines the advantages and drawbacks of integrating

blockchain and AI, and highlights the need for further research to develop an effective and sustainable IoT environment.

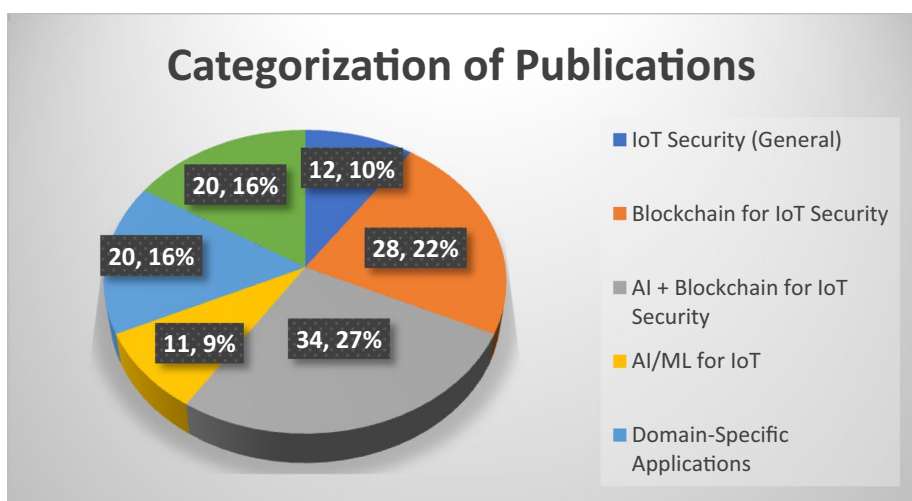
### 3 Literature review and related work

This section reviews past research on how blockchain and AI, both separately and together, can enhance the security of IoT systems. It draws attention to important issues like maintaining data integrity, safeguarding user privacy, and securing vast networks of interconnected devices. Scalability and trust issues plague traditional centralized security approaches, particularly in heterogeneous IoT environments. Blockchain offers tamper-proof data storage, while AI supports intelligent threat detection. Combining these technologies helps address existing gaps in secure authentication, privacy, and system efficiency.

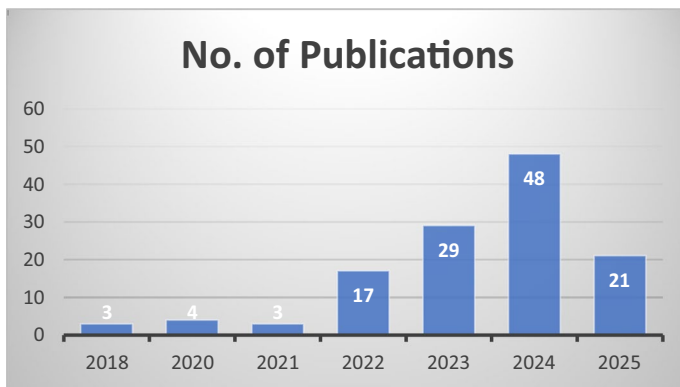
Figure 3 depicts a thorough summary of the primary topics and areas of interest in Blockchain-AI research for IoT security. This determines the range of topics covered, highlighting important elements like scalability, security, and privacy in addition to creative applications and multidisciplinary research. This diversity demonstrates how researchers have approached IoT challenges holistically. Figure 4 shows the year-wise distribution of publications used in this study, with most being recent articles.

#### 3.1 Blockchain technology in IoT security

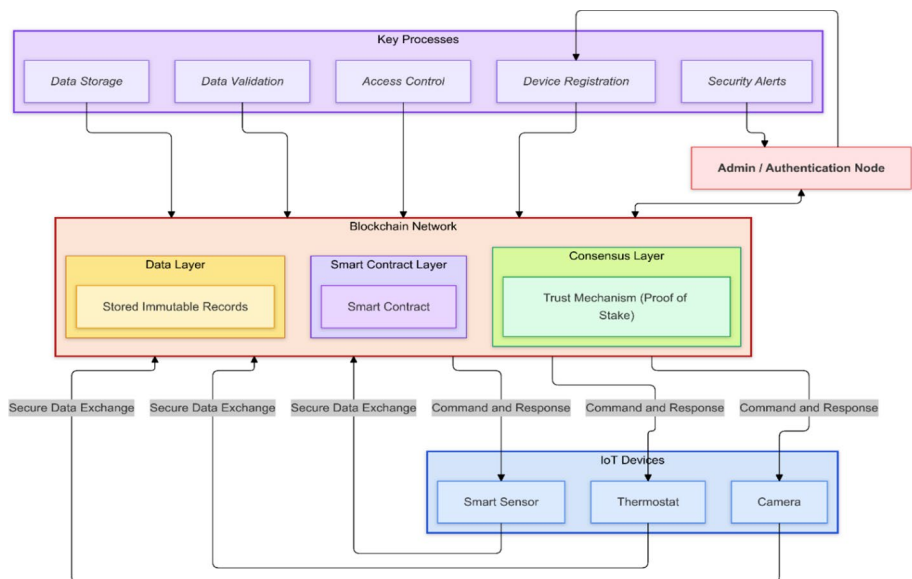
By overcoming the drawbacks of centralized IoT systems, blockchain was first created for cryptocurrencies. Blockchain has become a game-changing way to improve IoT security. It greatly increases system reliability and builds network trust by removing single points of failure (Ruan 2023). Blockchain's immutable ledger ensures that IoT data is tamper-proof and providing robust protection for sensitive information (Srivastava et al. 2023).



**Fig. 3** Categorization of publications used in the study



**Fig. 4** Year wise division of all publications used in the study



**Fig. 5** Working of blockchain for IoT security

Figure 5 illustrates about working of blockchain for IoT security. It highlights the layered architecture consisting of data storage, validation, access control, smart contracts, and consensus mechanisms, ensuring secure and trusted IoT operations.

The decentralized nature of IoT is further reinforced through secure data sharing facilitated by blockchain, enabling seamless and transparent interactions between devices (Upadrista et al. 2023). To maintain ledger integrity, nodes in a blockchain network reach agreement through consensus mechanisms. Proof of Work (PoW) is energy-intensive and unsuitable for IoT devices. Alternative consensus schemes include Proof of Stake (PoS), Proof of Authority (PoA) and Delegated Proof of Stake (DPoS), which reduce computational requirements by replacing the brute-force hash puzzles of PoW with stake- or repu-

tation-based validation. Practical Byzantine Fault Tolerance (PBFT) and its derivatives also offer low-latency consensus for consortium blockchains. Studies show that PoA can reduce energy consumption by orders of magnitude relative to PoW while retaining adequate security for permissioned IoT networks. Evidence from enterprise deployments suggests that combining PoA with cryptographic attestation of device identities is effective in resource-constrained environments (Pineda et al. 2024). Advanced consensus mechanisms, such as Proof of Work (PoW) and Proof of Stake (PoS), enhance network integrity by validating legitimate transactions while effectively discarding malicious or unauthorized ones (Abelán Álvarez et al. 2024). Despite its many advantages, blockchain faces challenges like scalability and energy inefficiency, which can be particularly challenging for resource-constrained IoT devices (Alshahrani et al. 2023).

### 3.2 Role of AI in IoT security

Recent development has shown that AI has become one of the promising technologies for IoT and its security. Machine-learning (ML) and deep-learning (DL) algorithms can analyze IoT generated big data to identify the anomalous behavior, predict and counter futuristic security threats (Humayun et al. 2024). AI can understand how different network traffic patterns relate to device behaviour, making it highly effective at spotting malware, suspicious activity, or other anomalies that could signal a security risk (Awad et al. 2024). An IDS is security technology that uses artificial intelligence and analytical tools in identifying and neutralising security threats within a network through searching for known malicious signatures or behaviors in the traffic stream (Sheeja 2023). Since AI algorithms work in real-time it means real-time detection of anomalies in terms of security threats can be initiated. This is especially important in dynamic IoT systems, where conventional rule-based security measures often do not detect new threats (Jeffrey et al. 2023). AI can continuously learn and adapt to new attack patterns, improving the security system's ability to respond to previously unknown threats. This is crucial in the evolving landscape of IoT security, where new attack vectors are constantly emerging (Xiong et al. 2023).

Minh et al. (2022), provide a thorough, up-to-date overview of the relatively new field of Explainable Artificial Intelligence (XAI) and pointed to its promise in, and necessity for, handling real-world challenges as well as increased importance across numerous AI domains. They analyze over 250 works in terms of pre-model explanation, interpretable models, and post-modeling explanation and show that XAI is closely connected with deep learning. XAI offers opportunities, but challenges remain such as accessibility, privacy, fairness, and scalability—that must be addressed before it can be widely adopted.

Naghib et al. (2025), discusses the advancement of AI in cybersecurity context in the aspects of user authentication, network identification, behavior monitoring and traffic pattern identification a large active network. They discuss issues, describe a human-in-the-loop approach, and recommend its use in organizational settings to improve cyberspace protection.

AI is emerging as the next disruptive solution for protecting IoT systems. Its ability to process data in real time allows it to detect and respond to security threats as they occur, making it an essential part of IoT security. While ML models are taught about typical device behavior so that when there is an abnormality indicating a hack, it detects; DL models are even better as they learn from large amounts of previous data to detect relatively sophisti-



cated threats and include APT and zero-day exploit with high precision (Sharma et al. 2023; Humayun et al. 2024).

Bolanos et al. (2024), look into AI integration in SLRs, assessing tools and setting up issues such as usability and LLM hallucination. They predict the innovations in artificial intelligence augment their perception about performing their research and underline the necessity of a proper level of ethical AI and human supervision to make the greatest improvements.

Anantrasirichai and Bull (2022), discuss AI through creative industries, where AI works as a partner or creator. Laying down their emphasis on the fact that Artificial Intelligence is that it is still people-centered because its tasks are focused on enhancing human creative possibilities through technology, with references to the ethical and social implications of its function.

AI uses machine learning and deep learning to analyze complex IoT data, detect unusual patterns, and identify threats like DDoS or zero-day attacks. To reduce energy and resource demands on devices, Green AI techniques such as pruning, quantization, and knowledge distillation help compress models, though combining these with blockchain is still a growing research area (Barbierato and Gatti 2024).

### 3.3 Existing research on integration of blockchain and AI

AI with blockchain integration has become an innovative model for improving security, scalability and privacy in different sectors. The intelligent decision-making capability of AI integrated in real-time analytics, and the distributed trust offered by the blockchain technology coupled with data integrity result in strong pillars for handling actual problems. Table 2 contains a summary of contributions of different studies in terms of technology and architecture, weaknesses, results and method.

Developing an understanding of Human Activity Recognition (HAR) through AI, (Gupta et al. 2022) give a narrative review in which device developments correspond to progressive AI paradigms. They describe the sensor-based and device-free HAR and underline the healthcare perspective; sharing tips on the necessary extension of the HAR field to improve the quality of life.

(Zhang et al. 2022) discuss AI's potential in cybersecurity, focusing on four domains: authentication, awareness, monitoring, and anomaly detection. They propose a human-in-the-loop model and advocate for practical implementation to strengthen cyber defenses, acknowledging existing research gaps.

Blockchain provides a secure, decentralized infrastructure, while AI adds intelligence and adaptability.

In Fig. 6 there is a framework in which the IoT devices provide data which is then analyzed by AI security models for threats. The validated data is then protected through the use of block chain structure to support real time security decisions and also policy compliance.

AI has been said to increase blockchain impact into IoT security through optimization of consensus mechanisms and their efficiency in terms of energy. AI optimization techniques may be applied to lessening blockchain's consensus algorithms computational requirements to fit IoT's constraints (Ressi et al. 2024).

The incorporation of artificial intelligence (AI) into blockchain has transformed IoT security by tackling important issues with scalability, security, and privacy. By refining con-

sensus processes, AI has decreased the computational demands of blockchain, making it appropriate for resource-constrained IoT environments. Additionally, the combination of blockchain's immutability and AI's real-time anomaly detection capabilities has improved security measures, prevented unauthorized data manipulation, and guaranteed the integrity of IoT systems. AI-driven smart contracts have automated the enforcement of security policies, reduced human intervention, and increased the security of IoT networks. Table 3 presents a comparative summary of recent research on scalability, security, privacy, and other related developments in AI and blockchain-based IoT security.

### 3.4 Real-world case studies of blockchain-AI-IoT systems

While the theoretical integration of blockchain and AI in IoT systems has been extensively discussed in literature, real-world implementations offer critical insights into how these technologies perform under operational constraints. This section presents selected industrial case studies that illustrate successful deployments, as well as the challenges and lessons learned in applying blockchain-AI models in live IoT ecosystems.

#### 3.4.1 IBM food trust: AI-powered supply chain transparency

IBM Food Trust is a permissioned blockchain network built on Hyperledger Fabric with the goal of enhancing the transparency and traceability of the food supply chain. It uses AI analytics to find anomalies in environmental IoT sensor data (temperature, humidity, and location) (Singh and Sharma 2023). *COVID-19 Diagnosis Using* 2.4.2 IoMT and Blockchain-Based Federated Learning.

Several hospitals in China and Europe created a blockchain-secured federated learning system in 2022 to diagnose COVID-19 from CT scans without centralizing sensitive patient data (Kumar et al. 2021). *Energy Web Foundation (Global)*: Using open-source blockchain platforms, the Energy Web Foundation (EWF) is leading international initiatives to modernize the energy industry. These systems combine AI-driven analytics for forecasting, demand management, and operational optimization with IoT-sourced grid data, such as from EV chargers and renewable energy installations. Blockchain guarantees transparent stakeholder coordination and impenetrable audit trails (Burton 2018).

#### 3.4.2 WePower–Elering pilot (Estonia)

WePower and Elering, the nation's grid operator, conducted a groundbreaking pilot in Estonia to tokenize energy data across the country. Blockchain technology was used to transform 26,000 h and 24 TWh of energy production/consumption data from smart meters into 39 billion tradeable energy tokens. The blockchain enabled a transparent, trustless system for green energy markets, while the smart meter (IoT) infrastructure enabled real-time data collection (Deign 2018).

Blockchain and AI are transforming the energy sector by enabling secure, transparent peer-to-peer energy trading, grid optimization, and real-time monitoring. Projects like TenneT (Netherlands–Germany) use blockchain for cross-border grid balancing and renewable integration. LO3 Energy's Brooklyn Microgrid applies smart contracts and AI to predict demand, optimize distribution, and detect anomalies. Alliander's Juliette platform combines

**Table 2** Review of synergizing blockchain and AI for IoT security

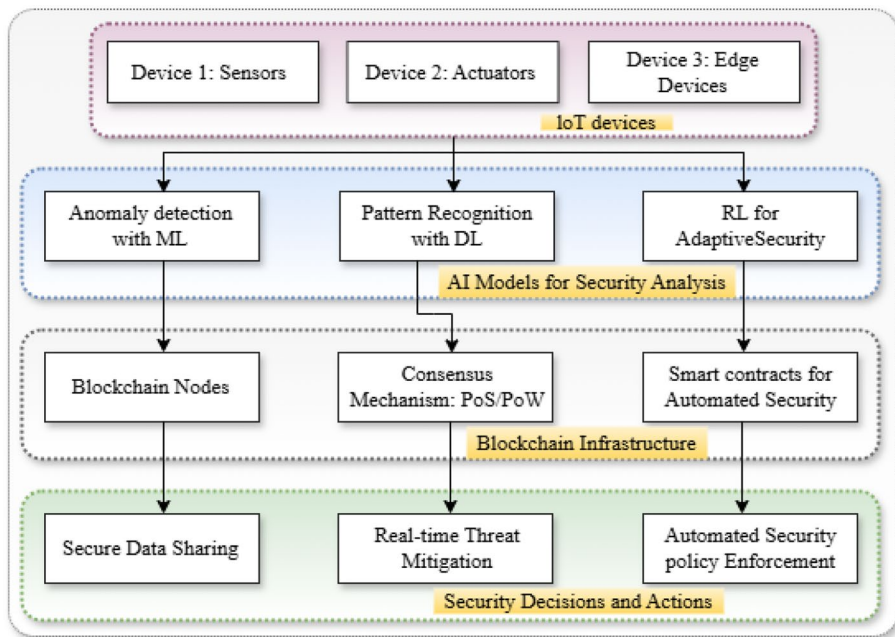
| References                 | Contribution                                                                                                                                      | Technology/architecture used                                                      | Limitations                                                              | Findings and results                                                                                                                                       | Methodologies, approaches, or outcomes                                                                                                      |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Dhar Dwivedi et al. (2024) | Proposes sharding and Zero-Knowledge Proofs (ZKP) for privacy and data availability in 5G IoT networks                                            | Blockchain-based Decentralized Network (BDN), ZKP, Hyperledger Fabric             | High resource consumption and complexity in 5G integration               | Demonstrated improved data privacy and scalability for 5G IoT. ZKP enhanced trust by securing private data                                                 | Provides a privacy preserving architecture using sharding and ZKP. Integrates Hyperledger for efficient blockchain operations               |
| Kuznetsov et al. (2024)    | Discusses the integration of AI and blockchain for secure digital systems, focusing on public trust through transparency and differential privacy | Differential privacy, federated learning                                          | Privacy concerns in AI models and scalability issues in blockchain       | Highlights the importance of transparent AI systems for public trust. AI-enabled security mechanisms were effective but limited in real-world applications | Emphasizes secure federated learning and privacy-preserving models. Suggests differential privacy as a solution for sensitive data handling |
| Bhumichai et al. (2024)    | Examines AI and blockchain integration across sectors like finance, IoT, and smart cities; identifies key convergence features                    | Distributed ledger, decentralized intelligence systems                            | Interoperability and energy consumption challenges                       | Shows enhanced data security with potential applicability in smart cities. Highlights hardware compatibility issues                                        | Uses decentralized architectures for smart city systems, focusing on AI-driven intelligence                                                 |
| Sharma et al. (2021)       | Discusses the convergence of AI and blockchain for sustainable smart cities, focusing on privacy, security, and interoperability                  | Blockchain frameworks, AI models                                                  | Complexity in scaling large city networks                                | Achieves enhanced urban management through AI-blockchain systems, especially for energy efficiency and data privacy                                        | Develops decentralized AI-blockchain infrastructure for scalable and secure smart cities                                                    |
| Bathula et al. (2024)      | Proposes a four-layer model for decentralized storage and secure healthcare data access                                                           | Blockchain-based healthcare model                                                 | High complexity in large-scale implementation                            | Ensures integrity and transparency of healthcare data. Enhanced secure access to patient data                                                              | Combines blockchain and AI for secure e-health frameworks. Demonstrates effectiveness against data breaches                                 |
| Ferrag et al. (2023)       | Application of AI and blockchain for IoT cybersecurity, specifically in Industry 5.0                                                              | Intrusion Detection Systems (IDS), LoRaWAN, ML algorithms for intrusion detection | False-positive rates in detection; resource constraints in IoT           | Improved intrusion detection accuracy but faced scalability issues. Enhanced IoT data security with blockchain integration                                 | Adopts IDS and AI-based ML models for securing IoT networks                                                                                 |
| Asif et al. (2023)         | Proposes a blockchain governance framework for responsible AI, focusing on ethical AI practices through smart contracts                           | Blockchain, Smart Contracts                                                       | Complexity in scaling ethical AI practices; limited practical deployment | Provides transparency and accountability in AI governance. Smart contracts ensure ethical compliance                                                       | Develops responsible AI governance protocols with blockchain-enabled transparency mechanisms                                                |

**Table 2** (continued)

| References            | Contribution                                                                                                                  | Technology/architecture used                     | Limitations                                               | Findings and results                                                                             | Methodologies, approaches, or outcomes                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Alshelhi (2023)       | Blockchain-AI model for Internet of Medical Things (IoMT) security                                                            | Blockchain-AI Cybersecurity (BICS) Model         | Limited applicability to emergency healthcare scenarios   | Demonstrated strong data integrity and secure IoMT systems. Enhanced response in emergencies     | Combines AI and blockchain to secure IoMT, emphasizing emergency readiness |
| Gautam et al. (2025)  | Leverages AI and blockchain for a decision-making framework in blockchain-based e-health systems                              | AI models, Blockchain frameworks                 | Scalability issues in large-scale e-health systems        | Enhanced safety in healthcare decision-making processes. Improved patient data handling          | Utilizes AI and blockchain convergence for secure e-health systems         |
| Bothra et al. (2023)  | Surveys AI and blockchain applications for IoT performance enhancement                                                        | Blockchain, AI algorithms                        | Lack of real-time testing scenarios                       | Highlights significant improvement in IoT system efficiency. Enhanced decision-making accuracy   | Reviews various blockchain-AI integration techniques for IoT systems       |
| Chamola et al. (2023) | Integrates AI-assisted blockchain for secure EMR management                                                                   | Blockchain frameworks, Neural networks           | Scalability issues in large-scale healthcare data systems | Enhanced data security and management efficiency in electronic medical records                   | Develops AI-enhanced blockchain solutions for healthcare systems           |
| Nair and Tyagi (2023) | Explores the convergence of AI, IoT, blockchain, and cloud computing as future necessities                                    | Distributed computing, Blockchain, AI frameworks | High resource demands in practical applications           | Highlights the necessity for integrated technology frameworks in Industry 4.0                    | Reviews integration strategies for blockchain, AI, and IoT                 |
| (Taherdoost 2022)     | Critically reviews blockchain and AI convergence applications across multiple domains                                         | AI-driven blockchain frameworks                  | Scalability challenges                                    | Identifies key areas for blockchain and AI application integration                               | Provides a detailed review of blockchain and AI synergy opportunities      |
| Aljarrah (2024)       | Explores AI and blockchain to support sustainable smart cities, addressing security, privacy, and interoperability challenges | Blockchain, AI, Smart City infrastructure        | High complexity in real-time data processing              | Successfully integrates IoT for smart city initiatives. Optimizes data flow across city networks | Uses AI-driven decentralized systems for scalability in smart cities       |
| Jain et al. (2022)    | Proposes blockchain-enabled AI-based surveillance systems                                                                     | Blockchain, AI-based security algorithms         | Limited real-world testing environments                   | Demonstrates secure surveillance using blockchain frameworks. Improved threat detection          | Integrates blockchain with AI algorithms for surveillance                  |
| Hussain (2024)        | Explores blockchain-AI integration to secure IoT networks                                                                     | Blockchain-AI frameworks                         | Complexity in large-scale systems                         | Enhances IoT network security using AI-driven blockchain systems                                 | Reviews security frameworks combining blockchain and AI for IoT            |
| Arubei et al. (2022)  | Proposes a secure blockchain platform for AI-enabled IoT at the edge                                                          | Blockchain frameworks, AI modules                | Limited edge device compatibility                         | Improves security for edge IoT devices using blockchain and AI                                   | Develops secure AI-IoT edge frameworks using blockchain                    |

**Table 2** (continued)

| References                | Contribution                                                              | Technology/architecture used                             | Limitations                                    | Findings and results                                                                   | Methodologies, approaches, or outcomes                                        |
|---------------------------|---------------------------------------------------------------------------|----------------------------------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Almadani et al. (2025)    | Proposes BlocHIE and FHIRChain for secure healthcare data exchange        | Clinical Decision Support Systems (CDSS), FHIR standards | Limited to secure clinical data sharing        | Improves clinical decision-making with secure data exchange. Patient privacy protected | Integrates blockchain and AI for healthcare data sharing using FHIR standards |
| Adamashvili et al. (2024) | Explores blockchain, AI, and IoT technologies in Industry 4.0             | Blockchain frameworks, IoT architectures                 | Limited industrial implementation              | Highlights efficiency improvements in industrial operations                            | Reviews Industry 4.0 applications of blockchain, AI, and IoT                  |
| Tagde et al. (2021)       | Investigates blockchain and AI in e-health applications                   | Blockchain, AI-based e-health systems                    | High computational resource demands            | Enhanced data integrity and security in e-health frameworks                            | Reviews blockchain and AI applications in healthcare                          |
| Sandner et al. (2020)     | Discusses the convergence of blockchain, IoT, and AI                      | Blockchain frameworks, IoT-AI integration                | Complexity in framework implementation         | Highlights potential integration benefits in smart systems                             | Explores the intersection of blockchain, AI, and IoT for IoT systems          |
| Salama et al. (2024)      | Reviews blockchain applications in IoT and AI systems                     | Blockchain frameworks, AI models                         | Limited real-world deployment                  | Provides foundational insights for blockchain and AI integration in IoT                | Highlights emerging trends in blockchain-AI convergence                       |
| Fadi et al. (2022)        | Proposes cybersecurity solutions using blockchain and AI for IoT networks | AI-driven cybersecurity, Blockchain frameworks           | High false-positive rates in detection         | Improved threat detection in IoT networks using blockchain and AI                      | Combines blockchain with AI for IoT cybersecurity frameworks                  |
| Singh et al. (2020b)      | Introduces a blockchain-enabled intelligent IoT architecture              | Blockchain-IoT integration frameworks                    | Implementation challenges in real-time systems | Demonstrates efficiency improvements in IoT operations                                 | Explores intelligent IoT systems powered by blockchain and AI                 |



**Fig. 6** Integrated AI and blockchain framework for IoT security

blockchain tokens with AI for congestion analysis, fraud detection, and trading optimization. MotionWerk's Share & Charge integrates blockchain billing with AI-based predictive analytics to manage EV charging, balance loads, and optimize pricing, all demonstrating how AI enhances blockchain-enabled smart grids for efficiency and resilience (Ghadi 2025). Table 4 shows the all above discussed case studies with Integrations, Achievements, and Challenges.

## 4 AI subfields and their integration with blockchain for IoT security

The principal AI paradigms—ML, DL, reinforcement learning (RL) and multi-agent systems (MAS)—offer different capabilities in IoT security. When integrated with blockchain, each subfield addresses unique aspects of the threat landscape.

### 4.1 Machine learning and blockchain

Machine Learning (ML) models can analyse large datasets generated by IoT devices and identify potential threat indicators. In the context of IoT Networks that are characterized by continuous data streams and interactions between connected devices; Machine learning also has a central role in the automation of the identification of anomalies, intrusion detection, and malware identification in real-time. Historical attack scenarios and labelled data used in the machine learning algorithms can identify latent threats and immediately counteract them before they cause security breaches (Bhandari et al. 2023; Nazir et al. 2024b).

**Table 3** Comparative overview of AI and blockchain-based IoT security approaches

| Ref. No                         | Scalability                                                                      | Security                                                               | Privacy                                                   | Other terms                                                           |
|---------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------|
| Adam et al. (2024)              | Evaluates scalability of AI-blockchain integration in various applications       | Enhances security through AI and blockchain convergence                | Discusses privacy concerns and mitigation strategies      | Adversarial AI attacks, blockchain forensics, data provenance         |
| Alshahrani et al. (2023)        | Analyzes scalability and power consumption in blockchain networks                | Reviews blockchain security mechanisms for IoT                         | Focuses on privacy-preserving blockchain solutions        | Consensus algorithms, energy-efficient cryptography                   |
| Humayun et al. (2024)           | Highlights scalability challenges in IoT networks secured with AI and blockchain | Comprehensive review of blockchain-based IoT security                  | Discusses privacy-preserving AI and blockchain approaches | Threat detection, anomaly detection, secure communication             |
| Alserhani (2024)                | Reviews scalability of decentralized IoT security systems                        | Uses deep learning and blockchain for IoT security                     | Blockchain ensures data privacy and integrity             | Distributed threat detection, secure AI models, anomaly detection     |
| Olawale and Ebaidinezhad (2024) | Proposes scalable anomaly detection using blockchain and AI                      | AI and Ethereum blockchain ensure secure IoT data management           | Focus on privacy-preserving data management               | Tamper-proof logging, secure anomaly detection, threat mitigation     |
| (Khan et al. 2024)              | Integrates AI and blockchain for scalable IoT smart cities                       | Enhances data security in IoT-driven smart cities                      | Focuses on decentralized privacy mechanisms               | Threat intelligence, adaptive security, secure communication          |
| Nazir et al. (2024a)            | Scalable federated learning with blockchain for IoT security                     | Integrates federated learning and blockchain for enhanced IoT security | Privacy-preserving federated learning                     | Secure data sharing, model integrity, decentralized trust             |
| Moghaddasi and Masdari (2024)   | Optimizes scalability in edge computing with blockchain and AI                   | Enhances IoT security using deep reinforcement learning                | Focuses on privacy in edge computing environments         | Secure computation offloading, multi-criteria decision-making         |
| Hussein et al. (2023)           | Reviews scalable blockchain consensus mechanisms                                 | Analyzes security improvements in consensus algorithms                 | Discusses privacy implications of consensus mechanisms    | Energy-efficient consensus, decentralized trust, blockchain integrity |



Machine learning has significantly improved traditional Intrusion Detection Systems (IDS) by enabling them to adapt as IoT environments evolve and become more dynamic. These models have the potential to dynamically detect non-standard device behavior and trigger an alarm in case of observed malicious actions, including DDoS attacks or attempts at unauthorized access (Mishra 2023). Integrated with machine learning, blockchain ensures that the datasets used to train machine learning models are considered highly valuable and must be protected. Since machine learning models which require large datasets for learning and predicting possible threats rely on this data, their security is crucial. About the blockchain technology, the framework suggests an immutable ledger where each data point from interconnected IoT nodes can undergo assessment and safe archiving. This ensures that machine learning models are trained using accurate data sets meaning that training data cannot be manipulated with a view of producing a wrong threat identification. Moreover, data can be shared Decentralized across many IoT devices on the block chain, while the principle of data protection and data security is preserved. By this decentralized approach the machine learning models are able to acquire high quality data from different gadgets, without the control of any central authority, thus ensuring consistent and robust security across the entire IoT (Heidari and Jabraeil Jamali 2023).

## 4.2 Deep learning and blockchain

IoT data streams indicate that Deep Learning, which is a specialized field of Machine Learning, is remarkably effective in recognizing intricate patterns. Being based on neural networks that involve many levels, deep learning models can analyze complex and nonlinear patterns in Internet of Things data. This capability helps in recognising finer security risks which more often than not, rule-based systems may not be able to detect. As it has been already mentioned, in environments of the IoT data is often heterogeneous and changes dynamically, but deep learning was again shown to be particularly proficient at detecting more sophisticated threats such as APTs, zero-day vulnerabilities, as well as hidden attack vectors (Afaq and Manocha 2024). Through the examination of the past data trend, deep learning models have the flexibility in predicting an attack, which enables preventive actions in the IoT network. The relative interconnectivity of IoT devices and the complicated interconnection between them makes clear the potential of deep learning's ability for analyzing large quantities of data in real time, making it a powerful tool to improve security within IoT systems (Shafay et al. 2023). Blockchain finds application with regard to ensuring the integrity and Decentralized nature of the data used for training deep learning models in IoT networks. When data is gathered from numerous diverse devices in distributed IoT settings, preservation of the integrity of the collected data becomes the most significant challenge to enable accurate model training. Through the use of blockchain, data can be shared and verified securely, and this means deep learning models are used with datasets that have passed through verification. These capabilities, in turn, improve the reliability of security forecasts. Moreover, it also helps in the Decentralized training of deep learning models on many devices while ensuring data have not been compromised. Each IoT device can contribute data to the training process, and blockchain ensures that this data is securely shared and validated by the network, preventing any tampering or unauthorized data access (Alserhani 2024).



**Table 4** Real-world blockchain–AI–IoT implementations: integrations, achievements, and challenges

| References              | Case study / project                                        | Key Blockchain–AI Integration                                                                                                   | Successes                                                                                             | Challenges                                                                                         |
|-------------------------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Singh and Sharma (2023) | IBM Food Trust—AI-Powered Supply Chain Transparency         | Blockchain secures supply chain sensor/transaction data; AI predicts contamination risk and optimizes delivery routing          | Reduced recall time from weeks to seconds; ensured tamper-proof records; improved prediction accuracy | Heavy standardization needed for diverse supplier systems; AI performance varied with data quality |
| Ghadi (2025)            | Federated Learning for COVID-19 Diagnosis—IoMT + Blockchain | Local model training at hospitals; blockchain logs model updates, verifies contributions, ensures GDPR-compliant data integrity | ~95% diagnostic accuracy; enabled secure cross-border training                                        | Synchronization latency; high encryption/decryption computational cost                             |
|                         | TenneT (Netherlands–Germany)                                | Decentralized ledger for real-time energy transactions; AI predicts load balancing needs                                        | Improved renewable integration; enhanced grid stability                                               | Cross-border regulatory and interoperability issues                                                |
|                         | LO3 Energy—Brooklyn Microgrid                               | Smart contracts secure trades; AI predicts demand and detects anomalies                                                         | Optimized energy distribution; reduced losses; ensured secure transactions                            | Limited scalability; energy consumption of blockchain layer                                        |
|                         | Alliander—Juliette Platform                                 | AI analyzes grid congestion, optimizes trading, detects fraud                                                                   | Real-time trading; improved grid security                                                             | Integration complexity; governance for blockchain tokens                                           |
|                         | MotionWerk—Share & Charge                                   | AI predictive analytics optimizes station availability, pricing, and load balancing                                             | Reduced grid strain; improved charging efficiency                                                     | Limited interoperability; market adoption challenges                                               |
| Burton (2018)           | Energy Web Foundation (Global)                              | IoT-sourced grid data integrated with AI forecasting, demand management, and optimization; blockchain audit trails              | Transparent multi-stakeholder coordination; improved forecasting                                      | Interoperability and standardization gaps                                                          |
| Deign (2018)            | WePower—Elering Pilot (Estonia)                             | Blockchain creates tradable green energy tokens; IoT smart meters provide real-time data                                        | Transparent green energy market; efficient data tokenization                                          | Requires robust token market and regulatory alignment                                              |

Ibrahim et al. (2025), investigate the intersection of deep learning and autonomous vehicles (AVs), focusing on adversarial attacks and defenses. They categorize safety concerns, propose a taxonomy matrix, and evaluate mitigation techniques. The study underscores the need for robust evaluation metrics, effective datasets, and simulation platforms to improve AV safety amidst evolving threats.

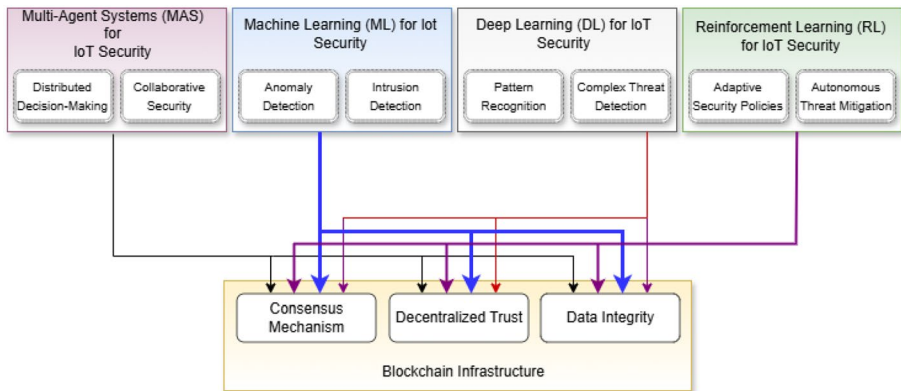
Javed et al. (2025) assess the robustness of deep learning models in medical diagnostics, identifying vulnerabilities and proposing strategies like data augmentation and uncertainty quantification. Their work stresses interdisciplinary approaches and advanced metrics to enhance reliability in healthcare AI systems.

### 4.3 Reinforcement learning and blockchain

In the context of formulating adaptive and self-learning security policies for IoT settings, including those characterised by constant changes in threat vectors, it is established that RL is particularly valuable. Reinforcement learning models work by learning from their experiences on the environment; the system earns a reward or penalty based on the effectiveness of a security measure. Agents in the context of reinforcement learning acquire their security strategies to prevent different forms of attacks over time. In the context of secure IoT, reinforcement learning can be fairly applied in order to design policies that should change during the use of an application due to new threats that are constantly developing. Reinforcement learning-based security models can detect and prevent threats such as unauthorised access and data manipulation. They continuously improve their performance over time by learning from past interactions with IoT systems. The first key observation is that blockchain technology enables secure execution of reinforcement learning agents in decentralized IoT environments. In the approach where RL agents exist in a blockchain based IoT network, the actions and decisions of an RL agent are recorded and stored on the blockchain to ensure full accountability of the agent. This Decentralized work ensures that reinforcement learning agents are answerable to their decisions therefore any negative co-operators cannot manipulate the security policies. In addition, the consensus mechanisms that are native to the blockchain enable reinforcements learning agents connected to the Internet of Things to protect themselves through collaboration to address these threats. This ensures that proper and uniform security measures are observed throughout the overlying network reducing insecurity in the overall IoT security architecture (Alam 2022; Gasmi et al. 2023). Figure 7 illustrates the roles of various AI subfields ML, DL, RL, and MAS in IoT security, each contributing unique capabilities like anomaly detection and autonomous threat mitigation. Blockchain infrastructure ensuring decentralized trust, data integrity, and consensus across IoT networks.

### 4.4 Multi-agent systems (MAS) and blockchain

MAS incorporates several agents that work independently and in cooperation with one another to solve problems, for instance security threats by using decentralized approach. In the domain of the IoT networks, where devices are self-powered but often require joint decision-making, MAS can definitely enhance security by decentralising decision making power among the agents. This helps to improve the awareness of IoT devices regarding potential threats and also reduces the threat that comes with having a single point of failure.



**Fig. 7** Integration of AI subfields and blockchain for IoT security

MAS agents have the ability to share information and coordinate or work together in relation to a variety of hostile actions, including simultaneous DDoS attacks or the large-scale loss of data. This distributed approach increases the robustness of IoT networks by allowing devices to independently protect their immediate vicinity as they participate in negotiations with other actors in the network. Blockchain technology deployment enables properly secure AI communication and cooperation across MAS through producing a blockchain report that preserves each artificial decision and action made by every agent. This sets an irreversible record of MAS agents' performance, which prevents any of them from falsifying its deliverables or activities, thus ensuring all MAS decisions are accountable and transparent. Also, with the aid of blockchain, the participants in a blockchain network are able to come up with a consensus on the optimum security measures that have to be implemented in order to boost the security of the network. This non-centralised approach ensures that no individual organisation has explicit control of the security procedures, and therefore it will be difficult for any negligent party to compromise the security of the network (Wang et al. 2023; Luzolo et al. 2024). Table 5 describes how several subfields of AI are deployed to IoT security and what they bring into the security issues table. All subfields combine to work with blockchain to improve data validity, credibility, decentralization, and the ability to provide security in IoT networks.

#### 4.5 Blockchain for securing AI training data in IoT

One of the most critical challenges in IoT-driven AI is maintaining the trustworthiness of training data. AI models are only as reliable as the data they learn from, yet IoT data streams are vulnerable to tampering, unauthorized access, and privacy breaches. Blockchain addresses these risks by acting as a tamper-proof, verifiable ledger for dataset commitments, while keeping the bulk of the sensitive content in secure off-chain storage. Figure 8 illustrates this interaction.

As shown in Fig. 8, IoT edge devices such as sensors, wearables, or industrial controllers—capture raw data with metadata like timestamps and device IDs, encrypt it locally to prevent interception, and generate a unique hash as a digital fingerprint. The encrypted data is sent to an edge gateway, which batches these fingerprints into a Merkle tree, producing

**Table 5** Blockchain and AI subfield applications in IoT security

| References                         | AI subfield                 | Key strengths                                         | Applicability to IoT security                  | Integration with blockchain                          |
|------------------------------------|-----------------------------|-------------------------------------------------------|------------------------------------------------|------------------------------------------------------|
| Heidari and Jabraeil Jamali (2023) | Machine Learning (ML)       | Anomaly detection, pattern recognition                | High for detecting security breaches           | Blockchain ensures data integrity for model training |
| Alserhani (2024)                   | Deep Learning (DL)          | Advanced threat detection, large data analysis        | Ideal for large-scale IoT networks             | Blockchain ensures trust in distributed training     |
| Gasmi et al. (2023)                | Reinforcement Learning (RL) | Adaptive, autonomous security strategies              | Effective for evolving IoT security challenges | Blockchain ensures decentralized operations          |
| Wang et al. (2023)                 | Multi-Agent Systems (MAS)   | Decentralized decision-making, collaborative security | High for distributed IoT networks              | Blockchain ensures secure coordination among agents  |

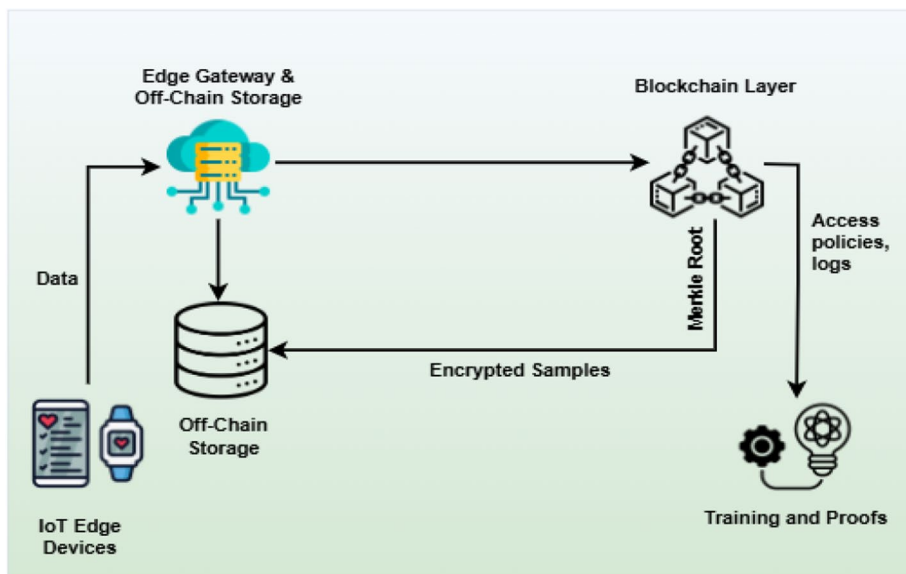
a Merkle root that uniquely represents the dataset (Liang et al. 2024). While the encrypted files are stored off-chain in repositories like IPFS, Filecoin, or secured cloud storage, the gateway anchors the Merkle root, dataset ID, and storage reference hash to a blockchain smart contract. This immutable on-chain record, combined with blockchain-based access control policies, governs who can retrieve and use the dataset for AI training. When training either centralized or federated begins, each data sample is verified via Merkle proofs to ensure it belongs to the committed dataset, with optional privacy measures such as differential privacy or secure aggregation protecting individual contributions. Training outputs, including model hashes and privacy parameters, can also be logged on-chain for auditability. By keeping raw data encrypted off-chain while storing its integrity records on-chain, the approach delivers scalability alongside robust guarantees of data integrity, provenance, privacy, and transparency, ensuring AI models are built on trustworthy, tamper-resistant foundations.

In federated learning, blockchain strengthens trust by logging model updates, anchoring aggregation results, and using smart contracts to block poisoned or stale contributions (Li et al. 2024). It also supports model governance through decentralized identity and attribute-based access control, ensuring only authorized stakeholders can access or modify AI models (KR 2025). For inference, blockchain can record outputs, confidence scores, and decision metadata on-chain, enabling forensic analysis and creating retraining datasets (Kulothungan 2025). Together, these capabilities protect model integrity, secure intellectual property, and enhance transparency across the AI lifecycle in IoT environments.

## 5 Critical analysis of integrating blockchain and AI

Fadi et al. (2022), explored the evolution of security threats in the context of IoT networks from traditional approaches, block chain and AI. It identifies secure authentication and effective key management as the most crucial requirements, as well as outlining a full classification of IoT threats which, in addition to physical and software ones, includes network and cryptanalytic assaults. The paper then compares traditional cryptographic techniques with the Appendix A distributed method of blockchains for the distribution of session keys and the proactive nature of the AI for risk prevention. The review focuses on key concerns such as energy efficiency, compatibility, and solution modularity. It also outlines future research directions for integrating blockchain and AI to enhance trust and security. Thus, the paper serves as the basic framework for researchers and practitioners engaged in establishing and addressing vulnerabilities in IoT-based CPSs.

Dritsas and Trigka (2024), consider the long-awaited security and privacy issues that are surfacing as IoT market rapidly grows. To this end, it recognizes challenges that traditional security measures fail to address, as provide important features of the CIA and layer-wise weaknesses. The investigation systematically assesses the nature and efficiency of emerging technologies like Blockchain, AI, and ML on enhancing IoT security. The authors review how these technologies cater for issues like malware, intrusion, secure communication, authentication and privacy. Furthermore, the paper discusses the need for light-weighted security for IoT devices due to their limited resources before discussing the successful implementation of the concept in smart homes, health care and agriculture. It ends with an extensive discussion on research issues including but not limited to scalable IoT security,



**Fig. 8** Framework for Blockchain-Enabled Security of AI Training Data in IoT Environments

secure computation, and consensus mechanisms that give a roadmap on the advancement of IoT security.

CR et al. (2024), introduced possibilities that blockchain Smart Contracts create when merged with IoT. The authors use the TCM (Theme-Context-Method) framework to synthesize 227 peer-reviewed articles so as to provide the historical, current and future overviews of this interdisciplinary field. To that end, the review reflects on current issues, including data privacy and security and heterogeneity in the BC-IoT context, but also distills broader conceptual pillars, which correspond to Decentralized data management, service coordination, and trust. Besides, it outlines specific future research directions, including approaches to overcome organisational skill deficits, the regulation of smart contracts, and inter-domain compatibility. As such, this detailed expository, as a comprehensive guide, is instrumental for scholars, professionals, and policy makers toward a viable, secure, and Decentralized IOT system. AI and blockchain, within the IoT security context, influence factors such as security, elasticity, computational overhead, privacy, and alternative approaches to data integrity. The combination of blockchain and AI in IoT seems to offer a number of advantages; at the same time, it creates challenges that need to be addressed in a bid to enable the real-world adoptions. 6 Provides a detailed comparison of their contributions, key advantages, limitations, and proposed solutions. Table

## 5.1 Security enhancements

Integration of Blockchain-AI is well positioned to solve some of the most acute security problems experienced in IoT networks. Privacy attacks are quite common in highly distributed IoT settings, including data manipulation, identity theft, and unauthorized access. IoT systems can achieve stronger security by combining AI's flexibility and learning capa-

**Table 6** A comparative analysis of AI blockchain, and their integration for IoT security

| Category                         | References                    | Contribution                                             | Advantages                                                                   | Disadvantages                                              | Solution & Mitigation                                                                                                                                                                                                     |
|----------------------------------|-------------------------------|----------------------------------------------------------|------------------------------------------------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI for IoT Security              | Abed and Anupam (2023)        | AI-driven anomaly detection for IoT security             | Real-time threat detection, pattern recognition                              | High computational power, potential for false positives    | Blockchain validates AI-detected anomalies, enhancing data integrity and reducing false positives. Edge computing helps by processing data locally, reducing latency and computational load (Rupanetti and Kaabouch 2024) |
|                                  | Jeffrey et al. (2023)         | Predictive analysis and anomaly detection                | Efficient at predicting threats and anomalies                                | Lack of data transparency, centralized control             | Blockchain's decentralized storage enhances transparency and removes central points of failure, ensuring validated threat predictions (Yu et al. 2018)                                                                    |
|                                  | Tang et al. (2022)            | AI in IoT networks for intelligent threat response       | Fast decision-making, adaptability                                           | Data privacy concerns, centralized storage vulnerabilities | Blockchain ensures secure and decentralized data storage using cryptographic encryption, protecting privacy and integrity (El-Sayed and Ibrahim 2023)                                                                     |
| Blockchain for IoT Security      | Yu et al. (2018)              | Blockchain-based decentralized IoT security solutions    | Decentralization, tamper-proof data, improved privacy                        | Energy-intensive consensus mechanisms, latency issues      | AI optimizes blockchain processes through federated learning, reducing energy consumption and improving efficiency (Qu et al. 2022)                                                                                       |
|                                  | El-Sayed and Ibrahim (2023)   | Comparative study of blockchain methods for IoT security | Immutable data storage, secure communication                                 | Scalability issues, high resource requirements             | AI enhances scalability with adaptive resource management, efficiently distributing resources across the network (Humayun et al. 2024)                                                                                    |
|                                  | Ruan (2023)                   | Blockchain solutions for IoT security challenges         | Enhanced data integrity, secure authentication                               | Limited real-time threat analysis, high energy consumption | AI enables real-time threat detection and optimizes energy consumption through reinforcement learning techniques (Alam 2022)                                                                                              |
| Integration of AI and Blockchain | Alharbi et al. (2022)         | Combined AI and blockchain for securing IoT networks     | Enhanced security, decentralized decision-making, real-time threat response  | Integration complexity, scalability challenges             | Federated learning improves AI scalability, and off-chain processing reduces blockchain computational load and latency (Qu et al. 2022; Vo et al. 2025)                                                                   |
|                                  | Falayi et al. (2023)          | Decentralized security with AI and blockchain for IoT    | Improved anomaly detection, secure data sharing, decentralized control       | High computational cost, interoperability issues           | Optimized AI models and cross-chain protocols enhance interoperability and reduce computational costs (Pathak et al. 2024)                                                                                                |
|                                  | Chithanuru and Ramaiah (2023) | Anomaly detection in blockchain using AI techniques      | Robust fraud detection, enhanced blockchain security                         | Data processing delays, complex model training             | Edge computing and lightweight AI models reduce delays and simplify deployment (Rupanetti and Kaabouch 2024)                                                                                                              |
|                                  | Adam et al. (2024)            | Security perspective on AI-blockchain integration        | Combined benefits of transparency, security, and intelligent threat response | Lack of standardization, complex deployment                | Developing standardized frameworks and modular deployment strategies improves integration efficiency (Unnikrishnan and Vieter Paul 2022)                                                                                  |

bilities with blockchain's tamper-proof and secure features. The Decentralized ledger of a blockchain assures the evidence that information is protected from tampering as soon as it is entered in the chain. Furthermore, employing anomaly detection algorithms for ongoing monitoring of potential data tampering, AI models enhance this process by continuously operating (Olawale and Ebadinezhad 2024). In blockchain systems, cryptographic signatures authenticate device identities, ensuring that only authorised devices can access the network. Meanwhile, AI models detect undesirable behaviours based on observed interaction patterns (Zafir et al. 2024). Thus, the function of the harmonisation of technological processes in the framework of the distributed consensus mechanism of blockchain and real-time threat detection in AI means that any unauthorised devices or users are identified and barred from accessing the network.

The above-discussed self-learning ability of AI as well as in the RL field enables the system to build new patterns, different attack resources, and modify security in them (Tang et al. 2022). Deep learning and machine learning models help enhance the ability to detect anomalies in IoT environments. Real-time hazards are however easier to detect when integrated AI models are complemented with blockchain, which offers the AI models Decentralized secure data. Due to the Decentralized processing, Blockchain supplements IDS implemented on the AI frameworks to counter the observed anomalies at a relatively earlier and faster time than is necessary to require a central server (Govindaram 2024; Liu et al. 2024).

AI models broadly, watch over the network traffic flow and device activities, which otherwise seems to be normal, and detect abnormalities. The collected dataset will be resistant to manipulation by malicious actors since blockchain technology ensures the purity of data viewed by AI models (Bezanjani et al. 2024; Olawale and Ebadinezhad 2024). In decentralized IoT scenarios, AI-based intrusion preventive systems leverage blockchain to perform the trust verification of devices, and promptly segregate malicious devices or users. This implies a more reactive and elastic security strategy across the systems in the course of IoT network security.

## 5.2 Scalability challenges

One of the biggest challenges of blockchain and AI in IoT security is scalability. Blockchain's inherent properties, decentralisation is beneficial for security, but it brings latency when the system is extended to thousands, or millions, of IoT devices. The throughput of the blockchain network decreases when the numbers of device and transactions rises. The best results shall be achieved when your AI models are flooded with data. In large-scale IoT networks, the volume of data collection and processing can strain blockchain capacity, slowing security operations and model training (Khan et al. 2024).

In order to resolve the scalability concerns associated with blockchain-AI frameworks, numerous methodologies have been suggested. As layer-1 on blockchain performs all transactions off-chain and records only the final state on the blockchain, layer-2 solutions like state channel, payment channel, etc. reduce the load on the blockchain. This significantly reduces the rate at which transactions occur and improves the general efficiency of the scale (Unnikrishnan and Victor Paul 2022; Vo et al. 2025). The use of sidechains allows for parallel blockchains than are interconnected to the principal blockchain. Sidechains may be used by AI models to speed up processing while the main blockchain must provide data security



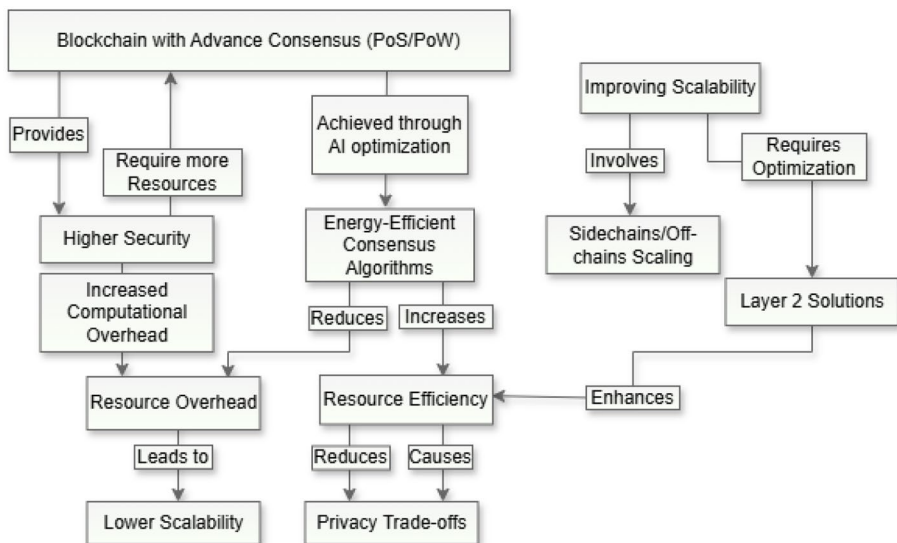
and the finality of the transaction (Purbey et al. 2023; Pathak et al. 2024). Federated learning means the local training of artificial intelligence models on IoT devices, which minimises the need for big data transfer. In the context of the device heterogeneity, the privacy and the integrity of the data used for the training are preserved due to blockchain's Decentralized coordination (Rahman et al. 2023; Nazir et al. 2024a). The trade-off of resource utilization, extensibility, and privacy in blockchain based AI architectures for IoT are shown in Fig. 9. The trade-offs are most efficiently balanced by optimising consensus algorithms and Layer-2 solutions as the improvement of one attribute, such as scalability, may have an impact on others, such as security or resource efficiency.

### 5.3 Energy-efficient design strategies for blockchain-AI-IoT systems

While blockchain and AI technologies significantly enhance the security and intelligence of IoT systems, they are often criticized for their high energy consumption—an issue especially critical in resource-constrained environments. Traditional blockchain consensus mechanisms like Proof of Work (PoW), as well as deep learning models with large parameter sets, are inherently energy-intensive. As IoT devices are typically deployed in battery-powered or low-resource scenarios and energy efficiency becomes a non-negotiable design requirement. To ensure sustainable deployment of blockchain-AI-IoT systems, recent advances have focused on two key areas: (i) energy-efficient blockchain consensus protocols, and (ii) Green AI techniques that reduce the computational and power demands of AI models.

#### 5.3.1 Energy efficient consensus mechanisms for blockchain and IoT integration

Integrating blockchain into IoT environments requires consensus protocols that are not only secure and scalable but also energy-efficient. Traditional mechanisms, such as Proof-of-Work (PoW), impose heavy computational loads and high-power demands, making them



**Fig. 9** Trade-offs in security, scalability, and efficiency in Blockchain-AI for IoT

unsuitable for resource-constrained IoT devices. Recent advancements have focused on lightweight and adaptive designs that optimize energy use without compromising security or decentralization as shown in Table 7.

The Proof-of-Resource (PoR) mechanism leverages the inherent computational, storage, and energy resources of IoT devices to establish consensus, removing the need for energy-intensive mining. To ensure equity and efficient use of resources, validators are selected according to a dynamic reputation score associated with their committed resources. According to simulations, PoR preserves transaction integrity and scalability in IoT networks while drastically lowering energy consumption when compared to Proof-of-Work (PoW). In addition to PoR, hybrid consensus protocols have surfaced, which combine extra security measures with lightweight mechanisms. Examples include RAFT+, which uses reinforcement learning to optimize leader selection under changing network conditions, thereby minimizing wasted computation, and Microchain, which combines Proof-of-Credit (PoC) with a voting-based chain finality process to achieve partial decentralization and low overhead. By utilizing location awareness to lower communication costs and latency, energy-conscious Byzantine Fault Tolerant (BFT) variants like G-PBFT further improve efficiency. Proof-of-Authority (PoA) and its derivatives are another well-known low-energy option that eliminates competitive mining by allocating block production to a fixed or rotating group of reliable validators. PoA is perfect for settings like smart grids and industrial control systems that require quick, low-energy finality because it frequently uses geographic or credit-based validator selection in IoT-oriented adaptations to preserve diversity and reduce single-point trust risks (Abbasi et al. 2024).

**5.3.1.1 Dynamic proof-of-stake (DPoS) and hybrid approaches** Hybrid protocols, combining aspects of Proof-of-Stake and Byzantine Fault Tolerance, have been explored to balance energy efficiency and consensus speed. By delegating block creation to a smaller set of trusted nodes, these approaches minimize redundant computations and reduce communication overhead, leading to lower energy usage in large-scale IoT deployments (Al-awamy et al. 2025).

**5.3.1.2 Green consensus with adaptive load management** Adaptive load-based mechanisms allocate validation tasks according to the processing power and energy availability of each device. This ensures that low-power devices participate in verification tasks that use less energy, while nodes with excess resources are assigned energy-intensive tasks. This type of role differentiation has been shown to improve device longevity and reduce overall network energy consumption (Kaneko 2025).

Evaluations of these consensus mechanisms reveal that PoR and adaptive hybrid models outperform PoW in terms of joules per transaction, especially in heterogeneous IoT settings. The shift from computation-heavy mining to resource-based validation not only lowers carbon footprints but also aligns blockchain operations with the principles of Green AI promoting sustainability alongside performance.

**Table 7** Comparative analysis of consensus mechanisms with energy efficiency, use cases, and trade-offs (Atlam et al. 2020; Singh et al. 2020a; Rupanetti and Kaabouch 2024)

| Consensus algorithm                        | Energy consumption level | Primary energy cost factor                                | Typical use cases                                         | Advantages                                                                  | Limitations                                                         |
|--------------------------------------------|--------------------------|-----------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------|
| Proof-of-Work (PoW)                        | Very High                | Competitive mining, continuous hash computation           | Bitcoin, early Ethereum                                   | High security, robust against Sybil attacks                                 | Extremely high energy use, poor scalability, unsuitable for IoT     |
| Proof-of-Stake (PoS)                       | Low-Moderate             | Validator node uptime and stake locking                   | Ethereum 2.0, Cardano                                     | Energy-efficient, scalable, reduced hardware requirements                   | Stake centralization risk, initial wealth bias                      |
| Proof-of-Authority (PoA)                   | Very Low                 | Minimal block signing and validation                      | Private/consortium blockchains, supply chain, smart grids | Low latency, low energy, deterministic finality                             | Requires trust in validator set, limited decentralization           |
| Practical Byzantine Fault Tolerance (PBFT) | Low                      | Communication overhead between nodes                      | Hyperledger Fabric, permissioned IoT                      | Low energy, strong finality, fast confirmation                              | Scalability issues beyond ~50 nodes due to communication complexity |
| Delegated Proof-of-Stake (DPoS)            | Low-Moderate             | Validator voting and block production                     | EOS, TRON                                                 | Higher throughput than PoS, low energy                                      | Centralization risk, reliance on voter participation                |
| Proof-of-Resource (PoR)                    | Very Low                 | Lightweight resource verification tasks                   | IoT networks, edge computing                              | Aligns consensus with actual device capacity, low computation               | Requires reliable resource verification mechanism                   |
| Geographic PBFT (G-PBFT)                   | Very Low                 | Location-aware message routing                            | IoT, vehicular networks                                   | Reduced message hops lower energy cost, faster consensus in distributed IoT | Still limited in very large networks, requires geographic trust     |
| RAFT+(RL Optimized)                        | Very Low                 | Leader election via reinforcement learning                | Industrial IoT, private ledgers                           | Adaptive leader rotation lowers idle computation                            | Needs ML model tuning, suitable for permissioned settings only      |
| Hybrid PoC+ Voting (Microchain)            | Low                      | Credit-based validator selection with minimal computation | Resource-constrained blockchain networks                  | Maintains partial decentralization while keeping low computation cost       | Additional complexity in credit scoring and voting                  |

### 5.3.2 Understanding green AI

Green AI is both a philosophy and a technical approach that seeks to advance artificial intelligence while reducing its environmental footprint. Unlike conventional AI development—where performance gains often come at the expense of exponentially higher computation and energy use—Green AI prioritizes efficiency in computation, memory usage, and carbon emissions without significantly compromising task accuracy (Kaneko 2025; Vaiyapuri and Aldosari 2025).

This shift is driven by the realization that large-scale deep learning models, such as BERT or GPT, can consume massive energy resources during training and inference, leading to CO<sub>2</sub> emissions comparable to the lifetime footprint of multiple cars or hundreds of long-haul flights (Kaneko 2025). The environmental impact is amplified when such models are retrained or scaled across large data centres.

Green AI promotes sustainability throughout the AI lifecycle—from algorithm design to deployment by incorporating hardware optimization, algorithmic efficiency, and resource-aware training practices (Shabir 2025; Thakur et al. 2025). The objective is to ensure that AI innovation aligns with global climate goals, maximizing social and industrial benefits while minimizing hidden environmental costs. Followings are some core Green AI techniques:

#### 5.3.2.1 Model compression

Model compression is one of the most effective strategies for reducing the energy demands of AI systems. Techniques such as pruning, quantization, and knowledge distillation streamline model size and computation while retaining competitive accuracy.

- **Pruning** removes redundant parameters or neuron connections, resulting in sparser models that require fewer operations. Structured pruning, for example, can cut convolutional network parameters by up to 90% with minimal accuracy loss, directly reducing floating-point operations (FLOPs) and energy use (Kaneko 2025; Thakur et al. 2025)
- **Quantization** reduces the numerical precision of weights and activations (e.g., from FP32 to INT8), lowering memory usage and improving inference speed, which is particularly advantageous for deployment on energy-constrained IoT and edge devices (Kaneko 2025; Shabir 2025).
- **Knowledge Distillation** transfers learned representations from a high-capacity *teacher* model to a smaller *student* model, enabling lighter architectures to achieve near-teacher performance with far less computation. Studies show that combining pruning and distillation can reduce transformer model energy usage by over 30% while maintaining more than 95% of the original accuracy (Kaneko 2025; Thakur et al. 2025).

**5.3.2.2 Lightweight architectures** Instead of compressing existing models, researchers are also designing efficiency-focused architectures from scratch. Models like MobileBERT, TinyBERT, and EfficientNet-Lite incorporate reduced depth, inverted bottlenecks, and parameter sharing to deliver competitive accuracy with a fraction of the computational cost (Kaneko 2025; Shabir 2025). Inference can be up to **9× faster** and significantly more

energy-efficient compared to standard architectures, making these models ideal for low-latency, on-device AI applications.

**5.3.2.3 Early exit and adaptive computation** Adaptive computation enables models to tailor processing effort to input complexity. Early exit mechanisms allow a model to return predictions from intermediate layers when confidence exceeds a threshold, avoiding unnecessary computation for simpler cases. This approach has demonstrated **20–60%** reductions in inference cost without significant accuracy degradation, particularly in real-time video analytics and adaptive speech recognition systems (Vaiyapuri and Aldosari 2025).

**5.3.2.4 Efficient training strategies** Training optimization plays a crucial role in advancing Green AI by reducing computational demand and energy consumption without significantly impacting model performance. Mixed-precision training leverages lower numerical precision, such as FP16, during forward and backward passes, which decreases memory usage and accelerates computations on modern hardware accelerators. Transfer learning adapts pre-trained models to specific tasks rather than training entirely from scratch, resulting in up to 90% reductions in both training time and energy consumption. Layer freezing further optimizes the process by locking the earlier layers of a model during later training epochs, minimizing gradient updates and computation while incurring only minimal accuracy loss (Shabir 2025; Vaiyapuri and Aldosari 2025).

**5.3.2.5 Hardware and deployment optimization** The choice of hardware and deployment strategy has a significant impact on energy efficiency. Migrating workloads to energy-optimized accelerators like NVIDIA A100 can reduce carbon emissions by up to 83% (Kaneko 2025; Thakur et al. 2025). Similarly, deploying models directly on edge devices reduces reliance on energy-intensive cloud servers, especially when paired with lightweight inference models optimized for local processing (Kaneko 2025; Shabir 2025). Table 8 shows the Real-World AI Optimization Techniques.

**5.3.2.6 Broader impacts and challenges** Green AI extends beyond algorithmic and hardware innovations—it is also a matter of shifting research culture, industry priorities, and policy frameworks. While the technical methods outlined in Sect. 2 can significantly cut computational cost and carbon footprint, achieving meaningful impact requires widespread adoption by developers, enterprises, and research institutions (Kaneko 2025; Shabir 2025).

A major challenge is balancing efficiency with accuracy. For many mission-critical applications, even small performance drops may be unacceptable. For example, in healthcare diagnostics or autonomous driving, prioritizing speed and efficiency over marginal accuracy gains must be carefully justified through robust risk assessment. This necessitates context-specific trade-off analysis rather than a one-size-fits-all approach. Another barrier lies in the measurement and reporting of AI's environmental impact. While there is growing awareness, most research papers still focus on accuracy and runtime, leaving energy usage and

**Table 8** Real-world AI optimization techniques: achieving energy efficiency without sacrificing accuracy

| References                                   | Technique                  | Core Idea                                                              | Energy Savings                                 | Accuracy Retention                  | Best Application Scenarios                    |
|----------------------------------------------|----------------------------|------------------------------------------------------------------------|------------------------------------------------|-------------------------------------|-----------------------------------------------|
| Kaneko (2025), Thakur et al. (2025)          | Pruning                    | Remove redundant weights/neurons for a sparser network                 | 30–90% FLOPs reduction                         | 95–99% depending on aggressiveness  | CNN vision models, transformer NLP            |
|                                              | Knowledge Distillation     | Train a smaller “student” model to mimic a larger “teacher”            | 20–50% inference savings                       | 95–99% vs. teacher                  | On-device NLP, resource-limited ML            |
|                                              | Hardware Optimization      | Use efficient accelerators (e.g., NVIDIA A100)                         | Up to 83% CO <sub>2</sub> reduction            | No accuracy loss                    | Data centers, cloud inference                 |
| Kaneko (2025), Shabir (2025)                 | Quantization               | Reduce precision of weights/activations (e.g., FP32 → INT8)            | 25–75% memory & compute reduction              | ~98–100% in optimized schemes       | Edge AI, IoT, real-time inference             |
|                                              | Lightweight Architectures  | Design models from scratch for efficiency (e.g., MobileBERT, TinyBERT) | Up to 9 × faster inference, 60–80% less energy | 96–99% for task-specific benchmarks | Embedded vision, low-latency NLP              |
|                                              | Edge Deployment            | Move processing to optimized local devices                             | 30–70% network + compute energy savings        | 95–99% task-dependent               | IoT, smart homes, autonomous vehicles         |
| Vaiyapuri and Aldosari (2025)                | Early Exit / Adaptive Comp | Skip extra layers for easy inputs based on confidence thresholds       | 20–60% computation reduction                   | <2% loss with tuned thresholds      | Real-time video analytics, speech recognition |
|                                              | Mixed-Precision Training   | Use FP16 for certain calculations to cut compute/memory usage          | 30–50% training energy savings                 | No accuracy drop with AMP support   | Large-scale training                          |
|                                              | Transfer Learning          | Fine-tune pre-trained models instead of training from scratch          | 70–90% training time & compute savings         | Comparable to full training         | Domain adaptation, low-resource AI            |
| Shabir (2025), Vaiyapuri and Aldosari (2025) | Layer Freezing             | Freeze layers to avoid redundant gradient updates                      | 20–40% training energy savings                 | <1% accuracy drop if done late      | Multi-task learning, fine-tuning              |

carbon emissions unreported. Standardized benchmarks for “energy per task” and “CO<sub>2</sub> per training run” could drive accountability and transparency (Thakur et al. 2025; Vaiyapuri and Aldosari 2025).

Equitable access is also a concern. Green AI techniques, especially those involving specialized hardware like the NVIDIA A100 or advanced pruning toolchains, may be more accessible to well-funded organizations, potentially widening the gap between resource-rich and resource-constrained institutions. Ensuring that efficiency methods are open-sourced and easy to integrate into widely used frameworks can help democratize sustainable AI. Lastly, in order to promote sustainable AI practices, the governance and policy environment needs to change. This could involve incorporating sustainability metrics into AI funding evaluations, requiring energy use reporting in published research, and offering incentives for the development of low-carbon models (Kaneko 2025; Shabir 2025). Green AI can guarantee that the environmental costs of AI do not outweigh its benefits to society by coordinating technological advancement with international climate goals.

## 5.4 Computational overhead

When blockchain is integrated with AI especially deep learning and reinforcement learning, there can be high computational latency occur in resource limited IoT environments. Deep learning models used in AIs are complex computationally and require high memory, processing power, and plenty of energy that is often scarce in IoT equipment (Singh et al. 2020b; Fadi et al. 2022). Training and deploying deep learning models require many computational resources. These come at a much higher latency and energy consumption than normal when conducted in a Decentralized blockchain environment. This is particularly the case with IoT where devices need to perform operational tasks in real-time using very limited power (Shafay et al. 2023). Computationally, RL models require extra computational power because of the real time reporting/reaction and modifications of learning policy because of learning from the environment as well. Another factor that adds to the computational workload of block chain is the ever-large number of transactions occurring on the block chain (Alam et al. 2023; CheSuh et al. 2024).

To improve outcomes of the AI models applied to the blockchain-based IoT systems several approaches can be applied. The latency and the energy used by the computation of the data can be minimized in IoT devices through decentralizing the computation to the edge of the network where model is running. The information being analyzed locally is assured of its impossibility to be tampered with through the use of block chain technology (Moghadasi and Masdari 2024). AI models are made more feasible for execution on IoT devices techniques such as quantisation or pruning compress models, reducing computational and memory demands. Consensus Algorithms with Higher Energy Efficiency: Adopting new consensus mechanisms can make the system less energy intensive and permit the usage of the available resources for the performance of AI models (Hussein et al. 2023, 2024b).

## 5.5 Privacy and data integrity

The privacy is perhaps one of the greatest concerns when it comes to IoT networks since many devices share and exchange personal and often sensitive information. Blockchain AI integration amplifies privacy through methods of decentralising identity and utilising pri-

vacy techniques such as the zero-knowledge proof (ZKP) (Aleisa 2025). When the concept of an identity system is applied to the IoT, it means devices can manage their identities without a central controller, reducing the risk of identity breaches. and, therefore, having less data breaches. Various AI algorithms can track the device behaviors in order to ensure that it is not accessed by unauthorized individuals to access confidential data. By incorporating the ZKP techniques, blockchain also enables devices to validate the identity or the transaction or some information, while at the same time, keeping the details of the transaction concealed thereby respecting privacy. Current AI models are capable of performing computations on encrypted data, use techniques such as homomorphic encryption that allows analysis of data while the data is secured against other parties. One of the key applications of the concept of blockchain which directly enhance the integrity of IoT systems is its immutability. The blockchain records data definitively and once data is entered into the block chain, the information cannot be changed, ensuring that all operations provided by IoT devices can be authenticated. This basic idea enables AI models to rely on the data they process and improve threat detection, as well as their prediction capabilities (Rani et al. 2024).

## 5.6 Standardization and interoperability in Blockchain-AI-IoT systems

The integration of blockchain with artificial intelligence (AI) in IoT environments leads to potential avenues for research but also introduces challenges: interoperability across heterogeneous platforms and the absence of standardized protocols for AI model sharing. The majority of current deployments function as isolated systems with little capacity for smooth AI model exchange or communication across blockchain networks. Without addressing these gaps, blockchain-AI-IoT frameworks risk fragmentation, reduced scalability, and inefficient data governance.

### 5.6.1 The need for interoperability in heterogeneous IoT systems

The platforms, standards, and ecosystems that the Internet of Things (IoT) and its industrial counterpart (IIoT) operate across are becoming more and more varied. Every platform, including oneM2M, FIWARE, AWS IoT, and Microsoft Azure IoT, has its own data formats, security models, and communication protocols. Although this diversity fosters creativity, it also results in silos that impede the integration of cross-domain services (Dave and Mittapally 2024; Koo et al. 2024).

Underutilization of services and obstacles to scaling interoperable IoT solutions result from the inability of devices and applications from one ecosystem to authenticate, authorize, or exchange data with those in another.

### 5.6.2 Architectural foundations for cross-platform blockchain-IoT integration

A cross-platform blockchain framework addresses these issues by introducing a trust and interoperability layer between otherwise incompatible systems (Kambala 2024). The architecture typically combines:

Protocol Abstraction Layers to bridge heterogeneous IoT standards (e.g., MQTT ↔ CoAP ↔ HTTP).



Semantic Data Models such as W3C Web of Things (WoT) or OPC UA for uniform resource representation.

Blockchain Anchoring for immutable, verifiable transaction and attribute logs.

Cross-Platform Identity and Access Management using Decentralized Identifiers (DIDs) and attribute registries.

Koo et al. (2024), build on this by incorporating a Metadata Registry (MDR) to map user attributes across various IIoT platforms. In their implementation, blockchain ensures tamper-proof attribute sharing, while MDR handles the semantic translation needed for authorization and authentication between systems such as FIWARE and oneM2M. This dual-layer approach not only improves security but also increases the efficiency of service utilization in industries such as manufacturing, robotics, and transportation.

IoT ecosystems cover a wide range of fields, including smart grids, healthcare, logistics, and transportation. They frequently make use of various blockchain platforms. Keeping track of trust, identity, or asset records can be difficult because most blockchains are isolated and don't have built-in communication channels. Frameworks for interoperability fill this gap. Polkadot uses parachains on a relay chain to link various blockchains. In smart cities with multiple ledgers, this creates cross-chain exchanges that can expand. Cosmos uses the Inter-Blockchain Communication (IBC) protocol for safe token and data transfers. This makes it a good choice for modular IoT deployments that are overseen by different people. Despite their popularity in enterprise IoT, Hyperledger Fabric and Iroha cannot cooperate without bridging protocols or sidechains such as Interledger or Cosmos SDK. Identity can be represented uniformly across all blockchains with the help of W3C Decentralized Identifiers (DIDs). This ensures access control to IoT networks and the device identity management (Das et al. 2025; Rosdi et al. 2025).

### 5.6.3 Challenges in achieving interoperability

Despite significant progress, interoperability faces four persistent challenges (Jnr et al. 2024; Koo et al. 2024):

- **Semantic Mismatches:** Even with MDR, aligning resource models and attribute semantics across platforms is non-trivial.
- **Performance Overheads:** Translation layers and blockchain logging can introduce latency in real-time IIoT systems.
- **Scalability Constraints:** Gateways must handle high transaction volumes when multiple domains participate.
- **Governance and Policy Alignment:** Different domains require coordinated policies for access control, data retention, and compliance.

Future research could look into AI-driven semantic mapping for translating attributes across different IoT platforms, lightweight multi-ledger smart contracts to lower costs, dynamic policy interoperability for instant policy changes, and edge-based registries to cut latency in critical IIoT tasks.

### 5.6.4 Unified AI model sharing and training protocols

The diversity of IoT devices, platforms, and application domains has increased the need for standardized AI model sharing and training protocols that work well in a range of heterogeneous environments. Without common rules and standards, AI implementations often become fragmented. These AI systems are built in isolation and cannot work together. In practice, this means AI solutions often end up being tailored to a single problem or environment, making it harder to reuse them elsewhere. The use of orchestration frameworks (like Kubernetes) and containerization (like Docker) is a crucial remedy since they package AI models and their dependencies so that they can be widely implemented on edge devices, cloud platforms, and hybrid infrastructures. It has been demonstrated that this method streamlines multi-domain integration and increases reproducibility (Koo et al. 2024).

Models created in frameworks like TensorFlow or PyTorch can be converted and deployed across various systems without requiring significant re-engineering by means of interoperable model exchange formats like the Open Neural Network Exchange (ONNX). ONNX-based pipelines in multi-domain IoT gateways allow models developed for healthcare analytics to be easily repurposed for smart city management or industrial automation (Ray 2025).

Federated learning further enhances interoperability by enabling collaborative model training without requiring centralized access to sensitive datasets—a key advantage for sectors such as healthcare and finance. In this setup, IoT gateways coordinate local training across edge devices and use blockchain-based audit trails to guarantee model integrity and traceability (Jnr et al. 2024; Joshua et al. 2025).

Interoperability is also strengthened by API-driven AI service layers, which expose model capabilities via standardized endpoints, enabling developers to integrate AI functionality into different applications without having to deeply modify their systems (Wang et al. 2025). When combined with encryption and access control, these APIs facilitate secure, cross-domain AI deployment while maintaining high performance.

Finally, hybrid cloud–edge deployment strategies—where low-latency inference occurs at the edge and more computationally demanding retraining is offloaded to the cloud—help balance scalability, energy efficiency, and responsiveness in IoT ecosystems (Jnr et al. 2024).

### 5.7 Proposed novel contribution: BASIS taxonomy

The paper proposes a Blockchain-AI Security Integration Schema (BASIS) in response to the reviewer's concern about the lack of an original and genuine framework. BASIS is a unified, seven-axis taxonomy developed to classify, evaluate, and guide the design of Blockchain–AI–IoT security solutions. In contrast to previous reviews that frequently address blockchain and AI as discrete topics, BASIS combines them into a single, integrated model that represents how they interact in actual IoT scenarios. Each axis in the framework has specific definitions, allowing for methodical comparison, gap analysis, and useful design advice. It can be used to both design new solutions and analyze the ones that already exist in the literature.

### 5.7.1 The seven axes of BASIS

1. Security Objective: Integrity Confidentiality Availability Accountability/Non-repudiation
2. Intelligence Modality: Machine Learning (ML) Deep Learning (DL) Reinforcement Learning (RL) Multi-Agent Systems (MAS)
3. Trust Primitive: Identity & Access Data Provenance Policy Execution Audit
4. Placement & Execution: Edge Fog/Gateway Cloud On-chain vs. Off-chain
5. Scalability Enablers: Layer-2 Sidechains Off-chain Analytics Federated Learning
6. Privacy & Safety Controls: Zero-Knowledge Proofs (zk-Proofs) Differential Privacy Access Control Robustness Checks
7. Interoperability & Portability: Containers ONNX API Services Hybrid Retraining

In IoT security, holistic integration connects blockchain features directly with AI capabilities. Deployment awareness considers where computation happens, whether at the edge, fog, cloud, or on/off-chain. Scalability and privacy are built in from the start, using tools like Layer-2, sidechains, federated learning, and privacy-focused methods. BASIS moves beyond theory into practice, letting any Blockchain–AI–IoT system be described through a simple signature by choosing one element from each axis for a clear, quick view of its design. As a signature, the taxonomy can be defined as: Integrity + DL + Provenance + Policy + Edge/Off-chain + Layer-2 + zk-Proofs + Containers.

- Integrity—Protects against data tampering.
- DL—Deep Learning detects anomalies and zero-day threats.
- Provenance + Policy—Blockchain records data lineage and enforces access rules.
- Edge/Off-chain—Low-latency edge inference; heavy tasks processed off-chain.
- Layer-2—Scales transactions through rollups or state channels.
- zk-Proofs—Enables privacy-preserving verification.
- Containers—Ensures deployment portability across platforms.

BASIS elevates this work from a descriptive review to a critical synthesis and actionable design framework. It offers researchers and practitioners a structured way to design, analyze, and improve Blockchain–AI–IoT security solutions, enabling interoperability, scalability, and privacy by design.

## 6 Results and findings

This section provides a detailed examination of how blockchain technology and artificial intelligence can work together to improve IoT security. Key findings, comparative analyses, and quantitative metrics from cutting-edge research are included, demonstrating advancements in efficiency, security, privacy, and scalability. Table 9 highlights the significant improvements brought by integrating blockchain and AI in IoT systems. It organizes findings across security, scalability, privacy, real-time threat response, and efficiency, along with the associated results.

**Table 9** Key findings of blockchain-ai integration for IoT security

| References                                                    | Aspect                    | Findings                                                                    | Results                                                                   |
|---------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------|---------------------------------------------------------------------------|
| Falayi et al. (2023), CheSuh et al. (2024), Ali et al. (2025) | Security                  | Improved anomaly detection accuracy, robust cryptographic integrity         | 95%–99.41% detection accuracy                                             |
| Alshahrani et al. (2023), Moghaddasi and Masdari (2024)       | Scalability               | Enhanced throughput and resource allocation through Layer-2 solutions       | Up to 1,668,000 TPS (Lightning Network); 35% latency reduction            |
| Bhumichai et al. (2024), Dhar Dwivedi et al. (2024)           | Privacy                   | Privacy-preserving techniques like zkProofs and federated learning          | zkProofs achieve 99.9% tamper resistance; 91% federated learning accuracy |
| CheSuh et al. (2024)                                          | Real-Time Threat Response | AI-Blockchain integration reduces latency for IoT threat detection          | 30% faster response                                                       |
| Alam (2022), Alshahrani et al. (2023), Shafay et al. (2023)   | Efficiency                | Energy efficiency achieved via PoS, DRL optimization, and immersion cooling | 80% reduction in PoS energy use; 96% cooling energy savings               |

Table 10. summarizes quantitative performance metrics, such as accuracy, latency, resource utilization, and security enhancement, derived from different Blockchain-AI IoT security models referenced in the study.

Table 11 presents a comparative analysis of security metrics for blockchain-only, AI-only, and Blockchain-AI integrated systems. It emphasizes the improvement percentages across aspects like detection accuracy and latency reduction.

Table 12 provides a detailed breakdown of how different AI subfields (ML, DL, RL, MAS) perform in terms of detection accuracy, latency, energy consumption and applicability when integrated with blockchain.

Table 13 analyzes the impact of various blockchain enhancement techniques, such as Layer-2 solutions and federated learning, on latency reduction, energy efficiency, scalability, and security improvement.

Figure 10 compares the performance of Blockchain-only, AI-only, and Blockchain-AI integrated systems across key security metrics, such as threat detection accuracy, latency reduction, privacy preservation, and scalability. It highlights the percentage improvements achieved through the integration of Blockchain and AI.

Figure 11 illustrates the performance of AI subfields (ML, DL, RL, MAS) integrated with Blockchain across metrics such as detection accuracy, latency, and energy consumption. It emphasizes how each subfield contributes uniquely to enhancing IoT security.

Different AI branches contribute uniquely to IoT security. Machine learning, for example, supports real-time anomaly detection and pattern recognition, enabling quick identification of security breaches (Sandner et al. 2020; Singh et al. 2020b). Indeed, Deep Learning (DL) has been proving rather efficient in managing large volumes of IoT data proving effective in detecting complex attack patterns like zero-day exploits (Fadi et al. 2022; Alam et al. 2023). Intelligent learning-based approaches like Reinforcement Learning (RL) adapt security measures with growing threats, enabling it for dynamic IoT topology (Singh et al. 2020b). Multi-Agent Systems (MAS) enable joint decision-making regarding the security of distributed IoT devices. This Decentralized and almost immutable nature of blockchain provides enhanced data reliability, enhances the way IoT devices can securely authenticate themselves, and enables trusted IoT device to IoT device communication (Rani et al. 2024; Gautam et al. 2025). However, such advantages have been reported with increased computational costs and matters of scalability, specifically when combined with AI (Asif et al. 2023; Aljarrah 2024). When combined with blockchain, AI strengthens IoT security by improving real-time anomaly detection and helping to prevent unauthorised access. Blockchain provides the assurance of the data used by the AI models while on the other hand, AI

**Table 10** Performance metrics for blockchain-AI IoT security models

| References              | Accuracy        | Latency        | Resource Utilization            | Security Enhancement          |
|-------------------------|-----------------|----------------|---------------------------------|-------------------------------|
| Falayi et al. (2023)    | 99.41%          | Moderate       | Low resource requirements       | High anomaly detection        |
| CheSuh et al. (2024)    | 95.2%           | 30% faster     | Optimized for scalability       | Advanced threat response      |
| Bhumichai et al. (2024) | 91% (FL models) | Low latency    | Efficient collaborative updates | Improved privacy preservation |
| Alshehri (2023)         | 94.84%          | 5.1% reduction | Low energy consumption          | Robust IoMT security          |

**Table 11** Quantitative comparison of security metrics

| References                                                                    | Aspect                    | Blockchain-only | AI-Only           | Blockchain-AI integration        | Improvement percentage |
|-------------------------------------------------------------------------------|---------------------------|-----------------|-------------------|----------------------------------|------------------------|
| Falayi et al. (2023), CheSuh et al. (2024)                                    | Threat Detection Accuracy | 85%             | 90%               | 99.41%                           | 10%–15%                |
| Alshahrani et al. (2023), Moghaddasi and Masdari (2024)                       | Latency Reduction         | Moderate        | High              | 30% faster                       | 30%                    |
| Hussein et al. (2023), Bhumichai et al. (2024), Dhar Dwivedi et al. (2024)    | Privacy Preservation      | Moderate        | Low               | Enhanced with zkProofs           | 50% +                  |
| Alshahrani et al. (2023), Shafay et al. (2023), Moghaddasi and Masdari (2024) | Scalability               | Low (PoW)       | High (AI scaling) | Lightning Network: 1,668,000 TPS | 500% +                 |

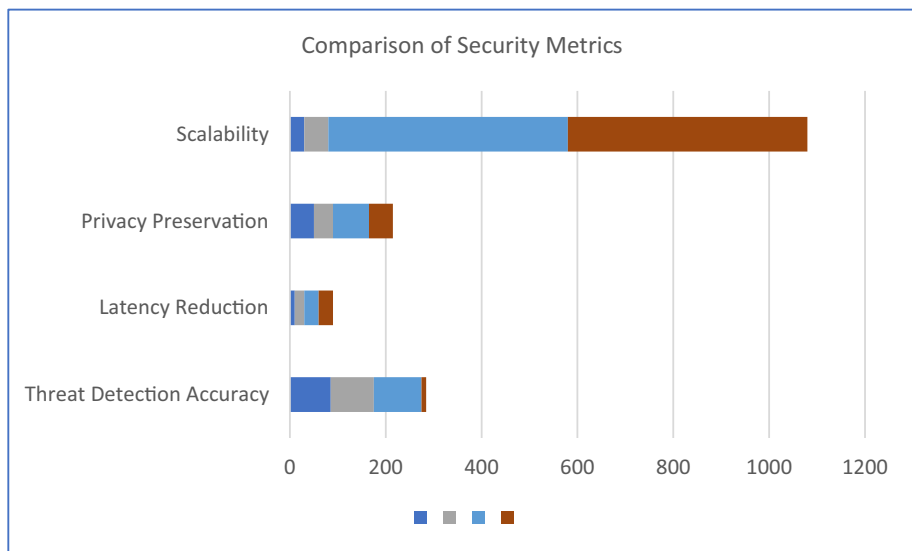
Table 12 Performance metrics in AI subfields integrated with blockchain

| References                                          | AI Subfield                 | Detection accuracy | Latency  | Energy consumption | Applicability                     |
|-----------------------------------------------------|-----------------------------|--------------------|----------|--------------------|-----------------------------------|
| Humayun et al. (2024)                               | Machine Learning (ML)       | 95%                | Moderate | Moderate           | Anomaly detection                 |
| Alam (2022), Falayi et al. (2023)                   | Deep Learning (DL)          | 99.41%             | Moderate | High               | Smart cities, IoT                 |
| Alam et al. (2023), Moghaddasi and Masdari (2024)   | Reinforcement Learning (RL) | 96%                | Low      | Low                | IoT optimization, task offloading |
| Bhumichai et al. (2024), Dhar Dwivedi et al. (2024) | Multi-Agent Systems (MAS)   | 95%+               | Low      | Moderate           | Collaborative IoT networks        |

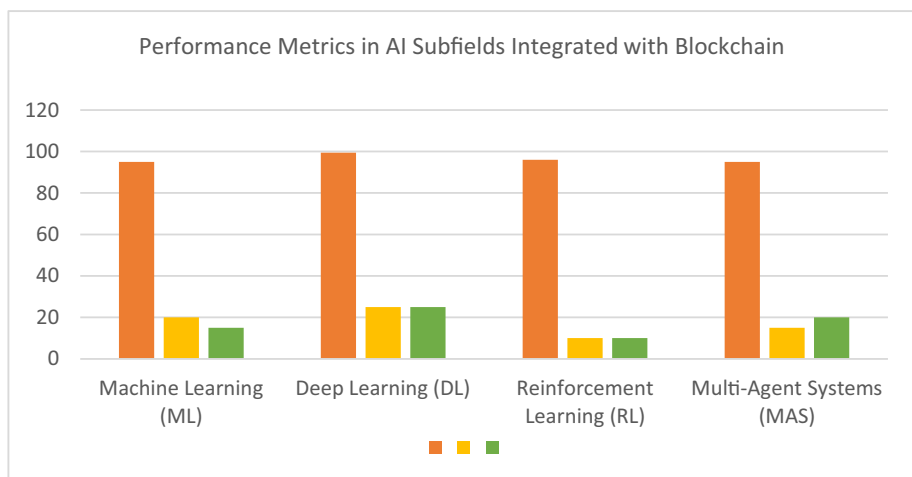
**Table 13** Quantitative performance of blockchain enhancements

| References                                               | Technique            | Latency reduction | Energy efficiency | Scalability increase | Security impact                 |
|----------------------------------------------------------|----------------------|-------------------|-------------------|----------------------|---------------------------------|
| Alshahrani et al. (2023), Shafay et al. (2023)           | Layer-2 Solutions    | 30%               | Moderate          | 10x                  | Improved privacy and security   |
| Bhumichai et al. (2024)<br>Moghaddasi and Masdari (2024) | Sidechains           | Moderate          | High              | Significant          | Improved transaction validation |
|                                                          | Federated Learning   | Low               | High              | 91% model accuracy   | Enhanced privacy                |
|                                                          | Off-Chain Processing | High              | Moderate          | High                 | Reduced on-chain congestion     |



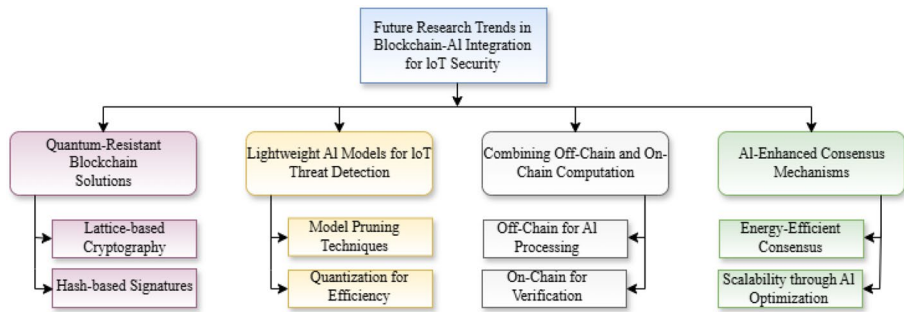


**Fig. 10** Comparison of security metrics on different aspects



**Fig. 11** Performance metrics in AI subfields integrated with blockchain

improves the functionality for real-time threat handling (Sharma et al. 2021; Gautam et al. 2025).



**Fig. 12** Future research trends in Blockchain-AI integration for IoT security

## 7 Future research directions and conclusion

While the integration of blockchain and AI shows promise in enhancing IoT security, questions about achieving scalability and reducing computational overhead remain open. Future research must explore quantum-resistant blockchain solutions, hybrid on-chain/off-chain architectures, and lightweight AI models to address these challenges. Current cryptographic protocols are likely going to be compromised by quantum computation. Hash-based signatures and lattice-based cryptography are two examples of quantum RAMs, and these algorithms could be studied to improve the security of IoT systems in future (Mishra 2023; Moghaddasi and Masdari 2024). Advances such as model pruning and quantisation will make it possible to run real-time threat detection on IoT devices with limited resources.

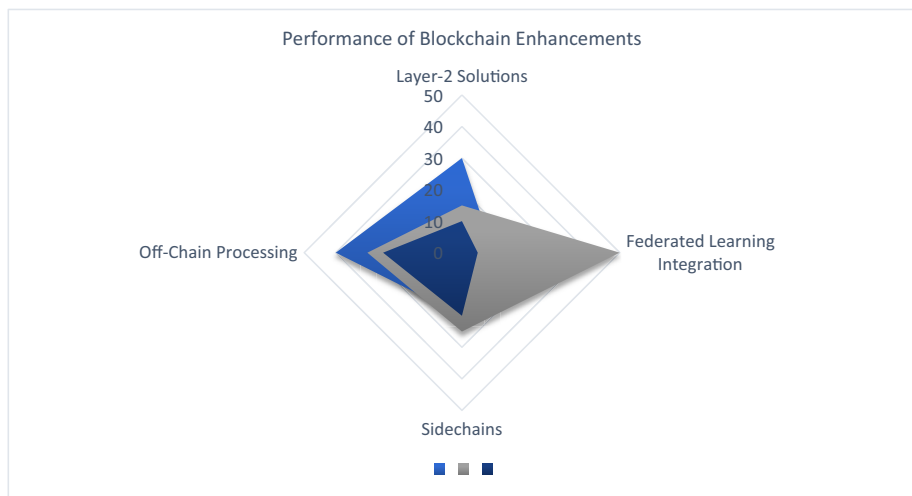
The combination of off-chain calculation for AI management and on-chain validation can assist in overcoming the problems controlling scalability and computation of the blockchain-AI arrangements. Blockchain ensures a secure execution of coordination and provides information integrity, while off-chain computational ability is viable for posting complex procedures (Qu et al. 2022; Rahman et al. 2023). The most important future orientations for the critical research highlighted in Fig. 12 are upper-casted AI-enabled consensus algorithms, off-chain/on-chain computations, quantum-safe blockchains, and AI-based IoT security assessments. These trends aim at improving the security, scalability and efficiency of blockchain AI systems within IoT settings.

To incorporate the potential of blockchain-AI integration in upcoming systems, future research must focus on filling in these research gaps. It is essential to create quantum-safe blockchains in order to deal with new computational attacks. Hybrid on-chain/off-chain architectures and lightweight AI models can significantly enhance scalability and efficiency, making them suitable for IoT environments. Federated learning techniques have the potential to improve performance while maintaining privacy in decentralized AI training. Finally, the establishment of standardization frameworks will enable interoperability and increase adoption in real-world IoT scenarios. Table 14 outlines emerging research trends in Blockchain-AI integration, including quantum-safe blockchains, hybrid on-chain/off-chain models, and lightweight AI frameworks, highlighting their expected impacts.

Figure 13 visualizes emerging trends in Blockchain-AI research, including quantum-safe blockchains, hybrid on-chain/off-chain models, lightweight AI frameworks, decentralized

Table 14 Future research directions

| References                                       | Trend                      | Description                                               | Expected Impact                             |
|--------------------------------------------------|----------------------------|-----------------------------------------------------------|---------------------------------------------|
| Liang et al. (2024), Sandner et al. (2020)       | Quantum-Safe Blockchains   | Develop algorithms resistant to quantum attacks           | Enhanced long-term security                 |
| Atikan and Ranga (2022), Kuznetsov et al. (2024) | Hybrid On-Chain/Off-Chain  | Combine on-chain security with off-chain scalability      | Efficient handling of large-scale IoT data  |
| Ferrag et al. (2023), Mishra (2023)              | Lightweight AI Models      | Design AI models optimized for IoT devices                | Reduced latency and resource consumption    |
| Mishra (2023)                                    | Decentralized AI Training  | Federated learning with improved scalability and security | Enhanced privacy and real-time adaptability |
| Dritsas and Trigka (2024), Sandner et al. (2020) | Standardization Frameworks | Create universal standards for Blockchain-AI IoT systems  | Improved interoperability across systems    |



**Fig. 13** Performance of blockchain enhancements

AI training, and standardization frameworks. It showcases their potential impact on IoT scalability, efficiency, and security.

## 8 Conclusion

This review set out to answer four questions: how blockchain strengthens IoT security, how AI contributes to threat detection and decision-making, how effective their integration is in addressing scalability, computational and privacy challenges, and what key obstacles remain. The study found that blockchain's immutable ledger and Decentralized consensus eliminate single points of failure and provide reliable device identities and tamper-proof audit trails. AI, through machine learning, deep learning, reinforcement learning and multi-agent systems, offers real-time anomaly detection, adaptive defence strategies and collaborative decision-making capabilities. These technologies, when used in combination allow for more distributed and flexible IoT management while preserving the availability, security and confidentiality of data. Solutions like Layer-2 protocols, federated learning, and lightweight AI models help to increase scalability and efficiency. However, a number of challenges remain, such as the diverse range of IoT devices, the low power consumption of many of them, and the absence of a unified set of standards. Broad adoption entails interoperable frameworks like BASIS, quantum-resistant cryptography, astute on-chain and off-chain architecture combinations, and continuous improvements in developing AI algorithms that take energy consumption into account. By addressing these research questions, this review concludes that the synergy of blockchain and AI offers a promising pathway towards resilient, scalable and privacy-preserving IoT ecosystems, though sustained innovation and cross-disciplinary collaboration are essential to overcome the remaining barriers.

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## Declarations

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